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Paleomagnetic study of the Late Silurian–Early Devonian Mt Daubeny Formation from the Broken Hill area, New South Wales

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The geological map of the Broken Hill area in New South Wales shows a striking feature, the Grasmere Knee Zone, which consists of a major change in structural trend. North of the Grasmere Knee Zone, the analysis of the structure of the Late Silurian–Early Devonian Mt Daubeny Basin coupled with AMS measurements suggests that the basin has undergone two phases of folding. Correction of magnetic data from bedding orientation has consisted in unfolding sequentially fold F2 to obtain a simple syncline and unfolding fold F1. Although the fold tests, conglomerate test and dyke test may be considered to be positive concerning the high-temperature component (DAU-C_H), paleomagnetic results from the Mt Daubeny Formation (locality DAU) are subject to caution, in particular due to the complex unfolding procedure. If component DAU-C_H, carried by hematite, is interpreted to be primary in origin, the corresponding paleopole is consistent with an X-type of apparent polar wander path for Gondwana, in particular if one relies on the proposed optimum bedding correction. South of the Grasmere Knee Zone, the Mt. Daubeny Formation is considered to be rotated clockwise relative to the north. The tentative model presented herein proposes that a block corresponding to the Southwestern Subprovince of Lachlan Orogen indented the Tasmanides between the Central Subprovince of the Lachlan Orogen and the Delamerian Orogen from the mid-Devonian (Tabberabberan event) up to the Early Carboniferous, triggering rotations in the Broken Hill area. A later magmatic event, thought to be Early Cretaceous, may have induced fluid migration and deposition of magnetite leading to the occurrence of an important magnetic overprint (DAU-C_M).

KEY WORDS: block rotation, Broken Hill area, Devonian, New South Wales, paleomagnetism, paleopole, Silurian, Tasmanides.

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INTRODUCTION

The tectonic setting of eastern Australia has been a matter of great debate over several decades. Briefly speaking, the Tasmanides of New South Wales are the result of three main orogenies: the mostly Cambrian Delamerian Orogen is observed in the west, the mid-Paleozoic Lachlan Orogen in the centre, and the Late Paleozoic–earliest Mesozoic New England Orogen in the east of the state. In the literature, two schools of thought have arisen as to whether the terranes (or zones) constituting the Tasmanides are parautochthonous, meaning that terranes underwent deformation but no substantial displacement or rotation compared with their relative present-day position (Schmidt & Morris 1977; Morel & Irving 1978; Goleby 1980; Schmidt *et al.* 1986,1987; Hargraves *et al.* 1987; Van Houten & Hargraves 1987; Van der Voo 1988; Offler *et al.* 1998), or whether the terranes are allochthonous, i.e. they were independent units that subsequently amalgamated along the margin of Gondwana (Embleton *et al.* 1974; McElhinny & Embleton 1974; Perroud *et al.* 1984; Veevers 1984; Livermore *et al.* 1985; Scotese *et al.* 1985; Leitch & Scheibner 1987; Powell *et al.* 1990; Glen *et al.* 1992; Gray *et al.* 1997).

In paleomagnetism, this controversy has led to the definition of two predominant apparent polar wander (APW) paths for Gondwana, of the many very different APW paths that have been published over more than 30 years (Morel & Irving 1978; Bachtadse & Briden 1990,1991; Kent & Van der Voo 1990; Schmidt *et al.* 1990; Klootwijk & Giddings 1993; Grunow 1999; Smith 1999; Li & Powell 2001; Torsvik & Van der Voo 2002; McElhinny *et al.* 2003). The basic difference between the two end-members was already highlighted by Morel & Irving (1978) who introduced the notion of an X- and a Y-type of APW path. The X-type considers that the terranes of eastern Australia are allochthonous and are not

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representative of the craton of Gondwana. By discarding Australian data, there is a gap in mid-Paleozoic paleomagnetic data and the X-type of APW path runs directly through Africa in a reconstruction of Gondwana from the mid-Ordovician to the mid-Carboniferous. On the contrary, the Y-type of APW path uses Australian mid-Paleozoic paleomagnetic data, implying that terranes of eastern Australia are parautochthonous. The addition of Australian data requires a Silurian–Devonian loop for the APW path passing through southern South America.

The publication of new results since 1978, in particular the Bokkeveld Group pole from South Africa (Bachtadse *et al.* 1987), and the Snowy River Volcanics pole from southeastern Australia (Schmidt *et al.* 1987), the Aïr Intrusive pole from Niger (Hargraves *et al.* 1987) and the Gilif Hills Volcanics poles from Sudan (Bachtadse & Briden 1991), merely accentuated the division of opinion rather than solved the problem. Currently, even the most recent APW paths published still reflect this discrepancy in the shape of the Gondwana APW path, whether they are those of Torsvik & Van der Voo (2002) or Smith (1999) who used numerical methods to smooth their path, or those of Grunow (1999) or Li & Powell (2001) and McElhinny *et al.* (2003) who tried to fit their path more directly to the data they selected. As an illustration of the mid-Paleozoic APW path issue, two paths, the X-type from Bachtadse & Briden (1991) and the Y-type of Schmidt *et al.* (1990), are shown in Figure 1.

REGIONAL GEOLOGICAL FRAMEWORK

The studied area is situated in far-western New South Wales, east and northeast of the city of Broken Hill (hereafter the 'Broken Hill area'). The Curnamona Province (or craton), a Paleoproterozoic succession deposited around 1700 Ma (Mills 1992; Scheibner & Basden 1996, 1998), extends from South Australia into New South Wales, and has been affected, in particular, by the Delamerian Orogeny around 510–505 Ma (Mills 1992, 2003; Scheibner & Basden 1996, 1998). The Broken Hill area, east of the Curnamona Province, is divided in three main zones from west to east: the Bancannia Trough, Wertago Zone and Kayrunnera Zone (Figure 2).

The Bancannia Trough adjoins the Curnamona Province along the Nundooka Creek Fault, corresponding for some workers to the Tasman Line (but see discussion in Mills, 1992 and Direen & Crawford 2003). The Bancannia Trough is a wide and deep north-northwest-trending graben bounded by the Nundooka Creek Fault on the west and the Bynguano Fault on the east. The trough is filled with up to 10 km of post-Delamerian Paleozoic siliciclastic sediments.

The Wertago Zone lies between the Bynguano Fault in the west and the Koonenberry Fault in the east. The Wertago Zone, from west to east, consists of Neoproterozoic shallow-water shelf sediments (Kara beds) containing a belt of subalkaline volcanics (the Mt Arrowsmith Volcanics: Crawford *et al.* 1997), Early to Middle Cambrian turbiditic slope deposits (Teltawongee beds), and

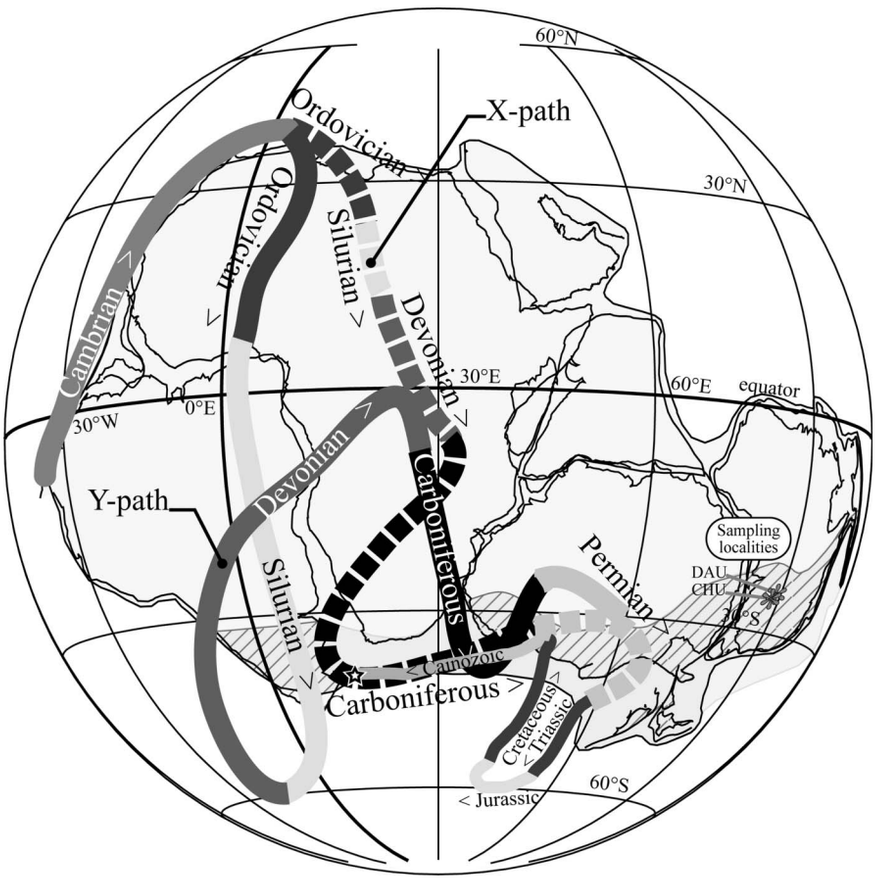


Figure 1 Two schools of thought on the shape of the APW path for Gondwanan-Australia. The X-type of path (dashed line), here after Bachtadse & Briden (1991), runs more or less directly through Africa during the mid-Paleozoic. The Y-type of path (solid line), after Schmidt *et al.* (1990), involves a Silurian–Devonian loop mainly based on paleomagnetic data from the Lachlan Orogen. The hatched zone represents the suspect terranes of Van der Voo (1993). Reconstruction of Gondwana in African coordinates; orthogonal projection centred at 07°S, 030°E.

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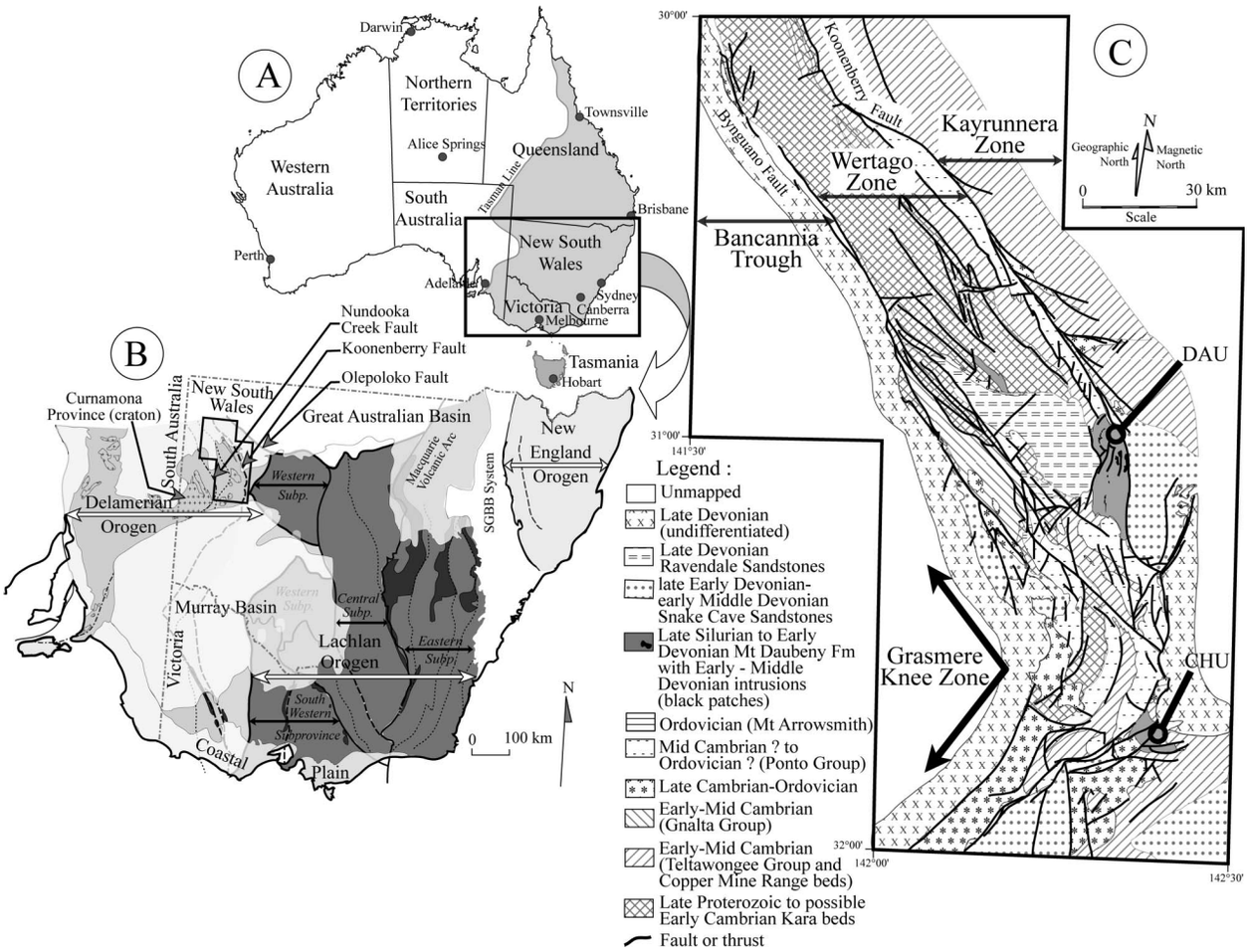


Figure 2 (a) The Tasmanides (grey area) of eastern Australia. (b) Main entities of the Tasmanides in New South Wales: the Delamerian Orogen, the Lachlan Orogen, and the New England Orogen. The Permian–Triassic Sydney–Gunnedah–Bowen Basins (SGBB) (white) covers the boundary between these two last orogens. In pale stipple are shown Mesozoic–Cenozoic basins, which cover most of the region. Dark zones represent Ordovician volcanic arcs, in particular the Macquarie Volcanic Arc. Boundaries drawn after Glen (2005). (c) Simplified geological map of pre-Permian rocks in the Broken Hill area, far western New South Wales, following Stevens *et al.* (2000). The bent double-arrow marks a change in structural trend, the Grasmere Knee Zone. The Mt Daubeny Formation (shaded grey) and localities DAU and CHU, where sampling has been carried out, are also indicated.

Early–Middle Cambrian distal turbiditic deposits with minor tholeiitic volcanics (Ponto Group). These units are generally bounded by Delamerian thrust surfaces and were strongly deformed at the end of the Middle Cambrian (Delamerian Orogeny). Less-deformed quartzose siliciclastic Late Cambrian (Mootwingee Group) and Late Silurian to Devonian sequences (Mt. Daubeny Formation, Snake Cave Sandstone and Ravendale Formation) unconformably overlie the older rocks. The Koonenberry Fault is a sharply defined brittle Carboniferous fault zone at the surface, but geophysical responses suggest that this fault may merge into a tectonic boundary of greater significance at depth (Mills & David 2004).

The Kayrunnera Zone lies between the Koonenberry Fault and the Olepoloko Fault to the north-east. The Olepoloko Fault has not been seen at the surface, being defined on the basis of airborne geophysical responses. The Kayrunnera Zone is mainly made up of tight upright folded turbidite facies sandstones of Early to Middle Cambrian age (Teltawongee beds).

Although the most important deformation events shifted to the Lachlan Orogen in the east during the Paleozoic, the Broken Hill area underwent major reactivation at this time. Gentle folding of Upper Cambrian rocks (Mootwingee Group, Kayrunnera Group and Cupala Creek Formation) in the Wertago and Kayrunnera Zones may be Late Ordovician to Early Devonian in age. Buckley (2003) suggested that dextral strike-slip movement occurred along the Koonenberry Fault in the Late Silurian to Early Devonian, leading to the opening of the Mt Daubeny Basin as a pullapart basin. However, this interpretation is refuted by Mills & David (2004), who believed that the Koonenberry Fault did not come into existence until the Carboniferous. In any case, the basin has been filled, primarily, with the non-marine lithic to siliciclastic Mt Daubeny Formation. Neef & Bottrill (1991) suggested a Lochkovian age (largely on the basis of plant fossils) for the formation, but the base, at least, is likely to be older. Indeed, there are andesitic to dacitic extrusive and pyroclastic units

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Table 1 Site mean directions and overall mean directions for locality DAU.

Name	*n*/*N*	*R*	D.InS	I.InS	D.Bed	I.Bed	α_{95}	κ	DipDir	Dip
Component DAU-C_M per site										
Dau-02	4/8	4.0	004.8	−67.5	047.5	−34.2	6.8	184.1	224.9	37.0
Dau-03	4/6	3.9	358.8	−66.9	058.1	−22.7	15.6	35.8	237.6	53.4
Dau-04	5/5	5.0	354.2	−73.6	049.8	−19.4	7.9	94.3	206.9	56.5
Dau-051/6 not used										
Dau-06	6/6	6.0	353.0	−64.6	062.9	−38.8	5.4	155.1	261.9	46.0
Dau-07	8/10	7.9	336.5	−60.7	057.8	−42.1	5.5	102.5	272.9	49.0
Dau-09	4/5	3.9	326.7	−70.3	040.9	−31.8	15.4	36.6	215.9	49.0
Component DAU-C_H per site										
Dau-02	8/8	7.8	025.4	−66.6	056.0	−30.5	10.1	31.0	224.9	37.0
Dau-03	6/6	5.9	007.9	−67.5	061.2	−20.6	10.6	40.7	237.6	53.4
Dau-04	5/5	4.8	328.2	−74.7	044.7	−24.7	17.1	20.9	206.9	56.5
Dau-05	6/6	5.9	338.3	−69.4	067.4	−45.6	7.5	81.2	262.9	46.0
Dau-06	5/6	4.9	330.3	−56.9	051.9	−44.5	14.0	30.7	261.9	46.0
Dau-07	9/10	8.9	000.6	−42.4	049.2	−28.8	6.7	60.1	272.9	49.0
Dau-09	4/5	3.8	038.4	−76.3	063.4	−27.3	25.9	13.5	215.9	49.0
Conglomerate test, component C_M (uniformity rejected at the 95% confidence level: negative test)										
Dau-01	10/17	6.8	349.9	−39.8	031.0	−17.5	35.6	2.8	244.9	46.0
Conglomerate test, component C_H (uniformity accepted at the 95% confidence level: positive test)										
Dau-01	12/17	4.9	046.2	−60.8	072.7	−15.9	57.3	1.5	244.9	46.0
Contact test (dykes), component C_M										
Dau-08 (dykes)	13/14	11.3	332.2	−72.0	∼	∼	7.2	16.7	Post-folding intrusions	
Dau-C_M-Sandstone	8/10	7.9	336.5	−60.7	057.8	−42.1	5.5	102.5	272.9	49.0
Dau-C_H-Sandstone	9/10	8.9	000.6	−42.4	049.2	−28.8	6.7	60.1	272.9	49.0

Name	*B*	*n*/*N*	*R*	D.InS	I.InS	α_{95}	κ	*R*	D.Bed	I.Bed	α_{95}	κ
Overall mean direction for component DAU-C_M												
Dau-C_M	6	31/46	6.0	348.8	−67.8	5.9	131.6	5.9	052.6	−31.7	9.4	52.1
Overall mean direction for component DAU-C_H												
Dau-C_H	7	43/46	6.8	356.6	−66.6	11.7	27.6	6.9	056.1	−31.9	8.9	47.0

B, number of sites; *n*, number of samples used out of *N*, total number of samples; α_{95}, radius of the cone calculated at the 95% confidence level; κ, precision parameter (Fisher 1953); *R*, resultant vector; D.Ins/I.InS, declination/inclination *in situ* (geographic coordinates); D.Bed/I.Bed, declination/inclination after optimum bedding correction; DipDir/Dip, direction of dip/dip for the bedding orientation.

The fold axis used for F1 is: 141°/12°; for F2: 249°/56°, until the bedding coincides with the general bedding of the limb taken as: 273°/30°.

In total, 9 sites & 77 cores, comprising 1 site & 17 cores in the conglomerate, 2 sites & 14 cores in the dykes, and 7 sites & 46 cores in the sandstone. Locality DAU situated at 30.9652°S, 142.6281°E.

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low in the formation that have been recently dated using SHRIMP zircon analysis at 425 ± 7 Ma (Black 2007). Furthermore, higher in the formation dacitic to rhyolitic sills are penecontemporaneous with the sediments and similar rocks occur as later cross-cutting dykes. Dating of these rocks gives a latest Silurian to earliest Devonian age [414.2 ± 2.7 Ma (Black 2005); 417.7 ± 1.9 Ma (Black 2006)]. In the late Early to early Middle Devonian, the basin was inverted and the Mt Daubeny Formation was folded. The general structure is a large syncline (F1), complicated by a fold (F2), which may be coeval with F1 or of subsequent generation. The Koonenberry Fault was clearly active in the Carboniferous and it has been suggested (Mills & David 2004) that dextral movements could be related to the Early Carboniferous intraplate Alice Springs Orogeny of central Australia.

The most striking feature on the geological map of the Broken Hill area (Stevens *et al*. 2000) is the Grasmere Knee Zone, a change in structural grain from north-northwest–south-southeast in the northern part of the map to northeast–southwest in the southern part (Figure 2). Two hypotheses have been put forward to explain this geometry. Either, the stress and strain

Table 2 Corresponding paleopoles for locality DAU.

Name	*B*	*n*/*N*	Australian coords		African coords		dp	dm
			Plong	Plat	Plong	Plat		
Paleopole corresponding to component C_H *in situ*								
Dau-M-InS	6	31/46	162.3	−68.6	043.5	−48.2	8.2	9.86
Paleopole corresponding to component C_H after bedding correction (100% unfolding)								
Dau-H-Bed	7	43/46	051.2	−37.6	026.1	12.2	5.7	10.9

B, number of sites; *n*, number of samples used out of *N*, total number of samples; Plong/Plat, paleopole longitude/latitude in Australian and African coordinates; dp/dm, semi-axis of the ellipse of confidence. Rotation parameters after Lawver & Scotese (1987).

Table 3 Site mean directions and overall mean directions for locality CHU.

Name	*n/N*	*R*	D.InS	I.InS	α_{95}	κ	*R*	D.Bed	I.Bed	α_{95}	κ
Component CHU-C_M per site											
Chu-01	5/5	4.9	000.6	−83.0	14.2	30.1	4.8	313.5	−09.0	18.8	17.4
Chu-02	6/6	4.8	282.3	−26.8	**38.2**	4.0	4.7	299.7	+49.4	**40.2**	3.7
Chu-03	4/5	3.9	188.2	+19.3	16.2	33.0	3.9	196.4	+17.5	17.0	30.3
Chu-04	6/6	5.8	007.6	−64.2	14.3	23.0	5.6	332.8	−12.6	20.6	11.5
Chu-05	6/6	4.3	323.5	−72.2	**47.5**	2.9	4.5	320.1	+03.1	**44.1**	3.3
Component CHU-C_H per site											
Chu-01	5/5	4.9	138.2	−60.6	18.3	26.3	3.9	277.0	−29.3	16.9	30.4
Chu-02	5/6	4.7	150.3	−60.0	20.6	14.8	4.7	284.9	−31.2	20.1	15.5
Chu-03	4/5	3.9	190.0	−48.2	19.9	22.3	3.9	262.9	+04.4	16.1	33.7
Chu-04	6/6	5.9	168.1	−49.5	10.5	54.2	4.9	264.4	−07.5	12.1	40.7
Chu-05	4/6	3.9	204.5	−52.3	18.2	26.5	3.9	272.0	+05.2	20.6	20.9

Name	*B*	*n/N*	*R*	D.InS	I.InS	α_{95}	κ	*R*	D.Bed	I.Bed	α_{95}	κ
Overall mean direction for component CHU-C_M (no real meaning)												
Chu-C_M	5	27/28	4.2	335.9	−59.8	**38.2**	5.0	4.1	329.4	+01.9	**40.1**	4.6
Overall mean direction for component CHU-C_H												
Chu-C_H	5	24/28	4.8	172.7	−56.5	15.7	24.7	4.8	271.7	−11.8	19.1	17.1

B, number of sites; *n*, number of samples used out of *N*, total number of samples; α_{95}, radius of the cone calculated at the 95% confidence level (when it exceeds 30°, values are shown in bold); κ, precision parameter (Fisher 1953); *R*, resultant vector; D.Ins/I.InS, declination/inclination *in situ* (geographic coordinates); D.Bed/I.Bed, declination/inclination after bedding correction.
Direction of dip/dip for the bedding orientation are not given, since it is variable within site; the fold axis used is: 139°/72°.
In total: 5 sites & 28 cores. Locality CHU situated at 31.7033°S, 142.7671°E.

fields were different in the north and in the south of the Broken Hill area, perhaps reflecting a compression against a strong and coherent buttress formed by the Broken Hill Block, or the southern part of the Broken Hill area has been rotated clockwise relative to the northern part. However, when and what caused this striking geological feature have remained unresolved.

SAMPLING AND PROCEDURES

Sampling

A paleomagnetic study has been carried out throughout the Broken Hill area, and this paper presents results from the Upper Silurian–Lower Devonian Mt Daubeny Formation, which is exposed both in the Mt Daubeny Basin north of the Grasmere Knee Zone (30.9652°S, 142.6281°E: locality DAU) and on Churinga Station south of the zone (31.7033°S, 142.7671°E: locality CHU) (Figure 2c). Outcrops of the Mt Daubeny Formation suitable for paleomagnetic purposes are rare in this desert region of low relief. The sampling localities had to be restricted to a few hundreds of metres of rocks exposed along seasonal creek beds. However, a reasonable number of cores have been collected (≥24: Van der Voo 1993) (Tables 1–4).

At locality DAU (Figure 2), the Mt Daubeny Formation consists of well-bedded (30–50 cm) fine-grained red-purple sandstones that overlie several beds of conglomerate. The conglomerates are made up of pebbles of rhyolites and grey sandstones derived from the Cambrian Gnalta Group, Ponto Group and Teltawongee beds. Three greenish-grey felsic (rhyodacitic) dykes intrude the top of the sampled section. Forty six

Table 4 Corresponding paleopoles for locality CHU.

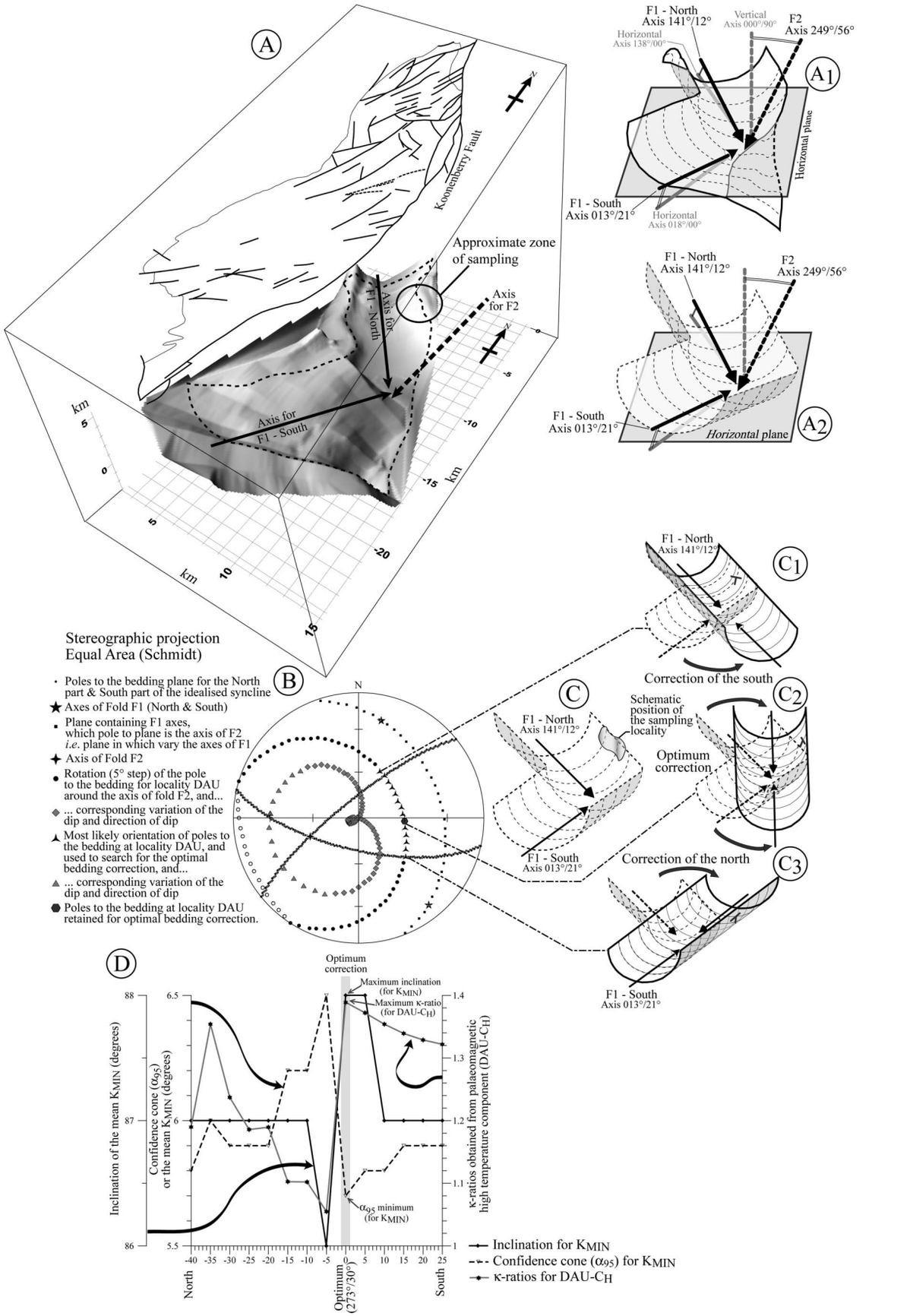
Name	*B*	*n/N*	Australian coords		African coords		dp	dm
			Plong	Plat	Plong	Plat		
Paleopole corresponding to component C_H *in situ*								
Chu-H-InS	5	24/28	136.4	21.1	127.2	07.8	16.3	22.6
Paleopole corresponding to component C_H after bedding correction								
Chu-H-Bed	5	24/28	228.5	−04.5	212.4	54.1	9.9	19.5

B, number of sites; *n*, number of samples used out of *N*, total number of samples; Plong/Plat, paleopole longitude/latitude in Australian and African coordinates; dp/dm, semi-axis of the ellipse of confidence. Rotation parameters after Lawver & Scotese (1987).

cores (seven sites) have been collected in the sandstones, 17 in the pebbles to perform a conglomerate test, and 14 in the dykes for a contact test. Although the Mt Daubeny Basin forms a large folded syncline (see below), the bedding orientation shows only small variations throughout locality DAU, implying that the different fold tests presented herein predominantly rely on site DAU-07, the most northerly in this locality.

At locality CHU (Figure 2), although having a different aspect probably reflecting different conditions of sedimentation, the fine-grained thin-bedded (5–10 cm) grey sandstones have been described as the stratigraphic equivalent of the rocks from the Mt Daubeny Basin (Stevens *et al.* 2000). The Mt Daubeny Formation locality sampled at Churinga Station is not well-exposed and forms a large syncline with a plunging axis, which can be regarded as cylindrical. Note, however, that the

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fold structure has to be considered totally independently from the structure described at locality DAU, which is some 80 km north of locality CHU. Metre-scale second-order folds, but with equivalent fold axes to the main syncline, affect the structure at locality CHU. Only one limb of the syncline has been drilled and 28 cores (five sites) have been collected. Because of these second-order folds, bedding orientation is variable within sites.

Field and laboratory procedures

The Mt Daubeny Formation was sampled using a gasoline-powered water-cooled portable drilling machine. Cores were oriented in the field using both magnetic and sun compasses. Data from the magnetic compass have been systematically used since no significant deviation has been observed from the sun compass. All orientations given herein are corrected from the local declination.

Standard cylindrical samples have been cut and measured in the laboratory at Munich. Anisotropy of magnetic susceptibility (AMS) was measured on one sample of every core using a Kappabridge KLY-2 (15 positions), and a sister sample subjected to stepwise thermal demagnetisation (one sample per core), since the AF technique, performed on pilot samples, failed complete demagnetisation. Natural remanent magnetisation measurements were performed with a 2G cryogenic magnetometer operating in the magnetically shielded room of the laboratory of the University of Munich, and interpreted with Palmag (Freeware, University of Munich). Rock magnetic measurements were carried out using a VFTB (Petersen's Instruments) and treated with the RockMag Analyser 1.0 software (Leonhardt 2006).

LOCALITY DAU: MT DAUBENY BASIN

Structure of the Mt Daubeny Basin and bedding correction

The 3D geometry of the Mt Daubeny Basin (Figure 3a) was inferred (and interpolated) from cross-sections published by Neef *et al.* (1989). In the north of the basin, the structure is a broad syncline, which transforms into a gently folded monocline to the south of the basin (Figure 3 A_1). Although relatively complex, therefore, the general structure of the basin can be reduced to a folded (or bent) syncline (Figure 3 A_2). A detailed analysis of the structural data obtained in the entire basin by Neef *et al.* (1989) shows a large syncline (F1), with an axis plunging to the southeast (141°/12°) in the northern part of the basin. The fold axis in the southern part plunges to the northeast (013°/21°). The change in fold axis direction of F1 between the northern and southern part of the basin corresponds to a second set of folds (F2) with a steeper fold axis (249°/56°). Sampling has been carried out where second-order folds exist, and the collected structural data are consistent with these being secondary F2 folds (Figure 3a). Both F1 and F2 folds can be considered as cylindrical folds although whether fold F2 is coeval with or later than fold F1 is undetermined.

The bedding correction therefore requires unfolding F2 in order to obtain a simple cylindrical syncline with a straight (linear but still inclined) fold axis for F1 throughout the basin, and unfolding and untilting F1 (the original bedding orientation is assumed to be horizontal). Hence, many bedding corrections are possible, because the original orientation of the axis of fold F1 is unknown. Given the position of the sampling locality in the basin, the pole to the bedding plane must lie on a small circle with a rotation axis of 249°/56° (i.e. the fold axis of F2: Figure 3b). However, it can be assumed that the correct solution (three branched stars in Figure 3b) is confined between a pole to the bedding plane corresponding to a fold axis F1 oriented 141°/12°, as in the northern part of the basin, and an axis oriented 013°/21°, as in the southern part (Figure 3c).

Sampling sites are located on only one limb of syncline F1 (Figure 3a, c) where the bedding attitude changes only slightly throughout the locality thanks to the second-order folds. Although the correction from the structure of the whole basin is of great importance for the final directional data, it has, therefore, little influence on the outcome of paleomagnetic fold tests (below).

Figure 3 (a) Block diagram showing the 3D geometry of the Mt Daubeny Basin (scale in kilometres). The shaded-grey surface is derived from interpolation of cross-sections published by Neef *et al.* (1989). The boundaries and main faults of the Mt Daubeny Basin (redrawn after Neef *et al.* 1989) are shown projected on the top of the block diagram for comparison (see Figure 2c). The general structure, shown as sketch in A_1 for sake of clarity, can be reduced to a folded syncline (A_2). The syncline (fold F1) has an axis plunging to the southeast in the northern part of basin, and plunging to the north-northeast in the southern part (solid black arrows). The syncline is refolded (or bent) by a fold F2 having a steep plunging axis to the west (dashed arrow). Second-order folds of F2 are also present, and their fold axes are considered to be equivalent to the main F2 axis. (b) Stereogram showing the solutions tested for bedding corrections. The axes of fold F1 in the northern and southern part of the basin (5 branched stars) are contained in a plane (squares), whose pole is the axis of fold F2 (4 branched star at intersection of the two great circles). The axis of F2 is also the intersection of the two great circles that are formed by the poles to all bedding planes (tiny dots) for an idealised syncline-like shape in the northern and southern part of the basin (the two end-member solutions). All possible poles to the bedding at locality DAU should lie in a plane, forming a small circle around the axis of F2 (circles), and possible solutions (3 branched stars) should be restricted between the two poles corresponding to the end-member solutions. The equivalent dip directions and dips are represented by grey diamonds and triangles. The optimum solution is shown by a hexagon. (c) Schematic illustration of corrections: C_1 & C_3 for the two end-member solutions, and C_2 for the optimum solution. (d) Graph showing the variation of inclination and α_{95} for K_{MIN} axes and k-ratio for the paleomagnetic component DAU-C_H with rotation around the fold axis of F2 (249.0°/55.5°). The optimum correction corresponds to a maximum in inclination value and k-ratio, and a minimum for the confidence cone (α_{95}).

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The question arises as to what bedding correction should be applied: rotating the southern part to align with the northern part (Figure 3 C_1); rotating the northern part to align with the southern part (Figure 3 C_3); or rotating both the northern and southern part to find an alignment in between the two end-members solution (Figure 3 C_2)? All possibilities have been tested every 5° steps, and a numerically best or optimum bedding correction has been found (Figure 3d).

The establishment of the optimum solution was first based on the analysis of the principal axes magnetic susceptibility (AMS), because the magnetic foliation in the sandstone of the Mt Daubeny Basin is clearly related to the sedimentary fabric (see below). Finding the optimum solution has consisted, therefore, in determining the rotation parameters for which the principal axis of minimum susceptibility (K_{MIN}) is the closest to the vertical (i.e. a magnetic foliation the closest to horizontal) and its ellipse of confidence is the smallest (meaning a better clustering of single sample K_{MIN}). The optimum solution has been found when fold F2 is corrected to coincide with a limb of F1 with a direction of dip and dip oriented 273°/30° (Figure 3d). The axis of fold F1 is then oriented 351°/08°.

The superiority of this correction is obviously tenuous. However, the fact that the independent dataset provided by paleomagnetic results yields a maximum κ-ratio for the high-temperature component using the

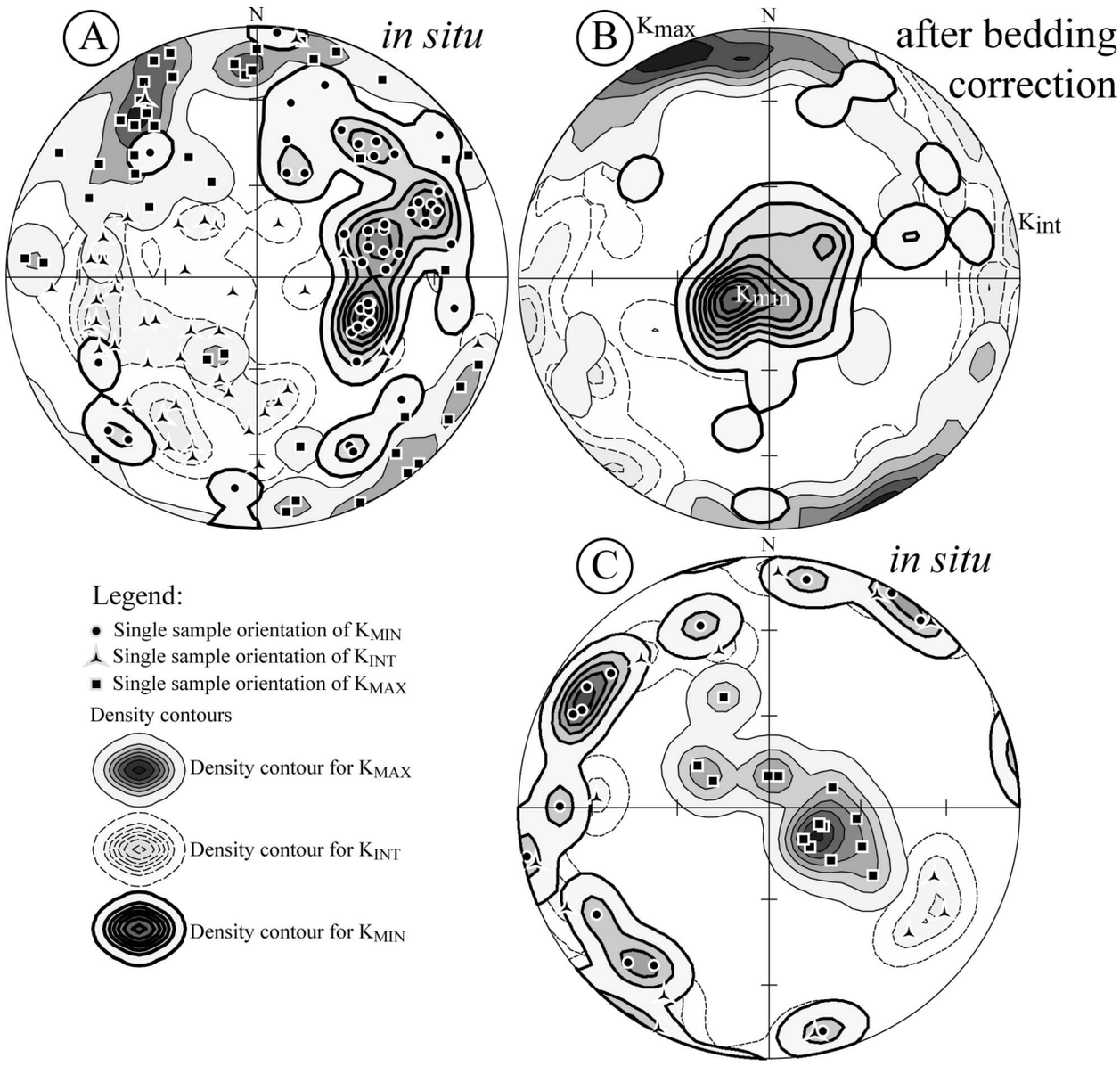


Figure 4 Orientations of the principal axes of magnetic susceptibility for the Mt Daubeny Formation (locality DAU), data points and density contours: (a) *in situ*; (b) after optimum bedding correction (for sake of clarity, only density contours are shown in b); and (c) for the dykes intruding the sandstone. K_{MAX}, squares and solid density contours; K_{INT}, triangles and dashed density contours; K_{MIN}, dots and thick density contours. Schmidt equal-area projection, lower hemisphere; Density contours calculated using a Gaussian model ($K = 100$).

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same bedding correction is a good support for this solution (Figure 3d). In addition, all structural directions and values of dip around 273°/30° have been scrutinised within 15° (i.e. the orientation has been varied in dip from 15° to 45° for all directions of dip between 258° and 288° every 1° steps), and the orientation of the limb of 273°/30° remains the optimum solution for both AMS and paleomagnetic data.

Finally, tests have been performed on the correction of AMS and paleomagnetic datasets when F1 and F2 are unfolded separately, and also from F1 and F2 are unfolded simultaneously. Fold tests performed on F1 or F2 unfolded separately does not improve or worsen significantly data clustering. It means that improvement in data clustering requires correction from both F1 and F2 folds. Concerning simultaneous unfolding (i.e. correction of 1° from fold F2 followed by 1° correction from F1 until complete correction), no clear differences have been found relative to a full correction of F2 followed by full correction of F1. Nevertheless, the magnetic lineation (K_{MAX}) obtained with a single phase of shortening is difficult to reconcile with the measured K_{MAX} axes given the present-day morphology of the basin.

Anisotropy of magnetic susceptibility (AMS)

The degree of anisotropy (Jelínek 1981) in the sandstones is very weak ($P_J = 1.02 \pm 0.01$) and the ellipsoid of

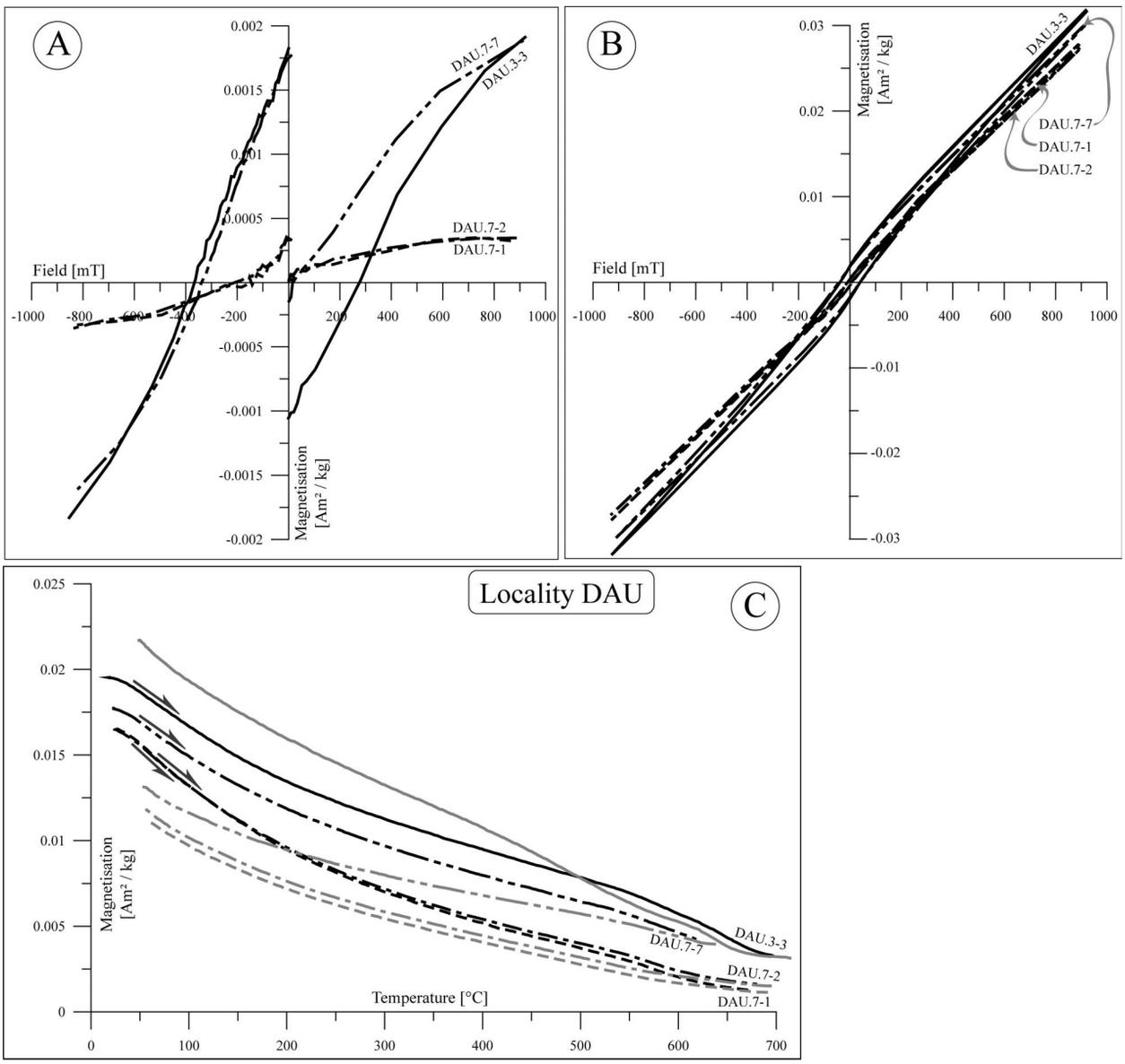


Figure 5 Examples of magnetic mineralogy measurements for locality DAU carried out with a Variable Frequency Translation Balance (Petersen Instruments) and analysed with the Rock Magnetic Analyser 1.0 (Leonhardt 2006). (a) Back-field (left) and IRM (right) curves; (b) hysteresis loop; and (c) thermomagnetic curves ($B = 530$ mT), with black lines corresponding to heating cycles, and grey lines to cooling cycles. DAU.3–3, solid line; DAU.7–1, dashed line; DAU.7–2, dotted-dashed line; DAU.7–7, double-dotted-dashed line.

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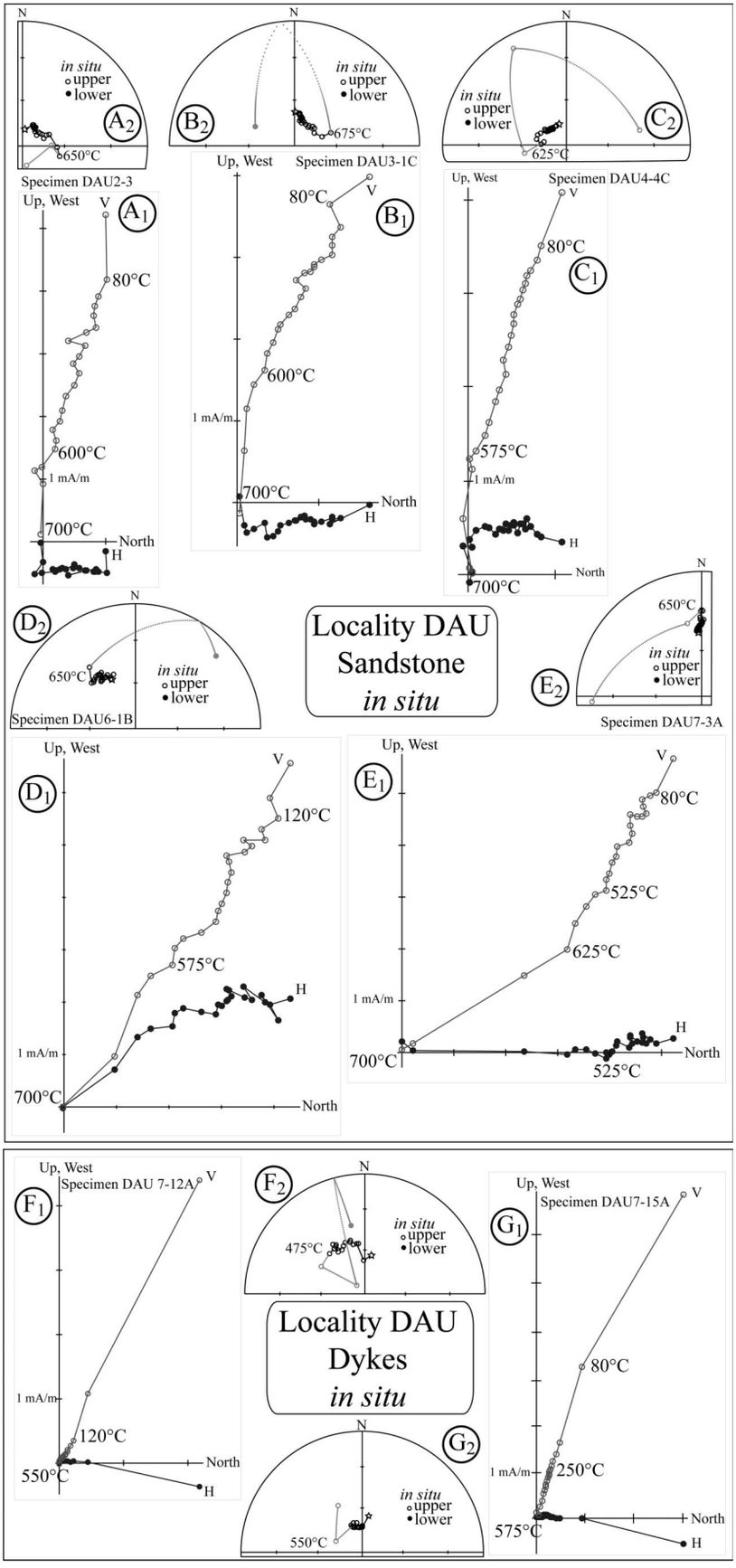


Figure 6 Examples of paleomagnetic results obtained by thermal demagnetisation from the sandstone of the Mt Daubeny Formation (locality DAU). (a) Specimen DAU.2–3; NRM at room temperature, 5.320 mA/m: A_1, orthogonal projection of the resultant vector *in situ* during demagnetisation, with close circles corresponding to the projection on the horizontal plane [H], and open circles, projection on the vertical plane [V]; $A_{2,}$, equal-area polar projection of the *in situ* orientations of the resultant vector of magnetisation during heating; the NRM measured at room temperature is shown with a star; coherent magnetic signal is drawn in black, while it is represented in grey when it becomes unstable; open circles are negative inclination values (upper hemisphere); closed circles, positive inclination values (lower hemisphere). (b) DAU.3–1; NRM at room temperature: 4.296 mA/m. (c) DAU.4–4; NRM at room temperature: 4.313 mA/m. (d) DAU.6–1; NRM at room temperature: 8.112 mA/m. (e) DAU.7–3; NRM at room temperature: 7.754 mA/m. In the intrusive dyke: (f) DAU.7–12; NRM at room temperature: 4.907 mA/m. (g) DAU.7–15; NRM at room temperature: 7.501 mA/m.

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magnetic susceptibility looks slightly oblate (shape parameter $T = 0.26 \pm 0.41$: Jelínek 1981; Hrouda 1982), although the standard deviation is large relative to the mean value. Moreover, the test of Parés & Van der Pluijm (2002) shows a degree of alignment calculated for phyllosilicates always lower than 10% (mean = 2.7% ± 1.8%). It is clear, therefore, that no significant internal strain, capable of deviating the remanent magnetisation, affects these rocks (Cogné 1987). The Königsberger ratio is about 1 ($Q_K = 0.85 \pm 0.30$: Königsberger 1938) and lightning effects are absent.

In the dykes, specimens show also a weak degree of anisotropy ($P_J = 1.06 \pm 0.02$) and the shape of the ellipsoid of magnetic susceptibility is close to a sphere on average as the shape parameter is close to zero ($T = -0.02 \pm 0.34$).

The directions of the principal axes of AMS (Figure 4) show clearly that a magnetic fabric related to sediment compaction has been preserved in the sandstone, because K_{MIN} axes are related to the bedding orientation, and the mean K_{MIN} axis is positioned about the vertical and K_{INT} & K_{MAX} axes about the horizontal after full (optimum) bedding correction (Figure 4b). Bootstrap analyses of AMS data (Tauxe 1998) show that K_{MIN} axes are better grouped ($\alpha_{95} = 5.7°$) when the optimum correction is applied. The direction of the magnetic lineation is then 337°/05° and has also its smallest confidence ellipse ($\alpha_{95} = 7.2°$). This direction corresponds fairly well with the direction of the fold axis F1, which has been rotated to N351° for optimum bedding correction.

The possibility that the magnetic lineation reflects the sedimentary flow at the particular place where sampling was carried out is dismissed, because the general flow direction reported by Neef & Bottrill (1991) is rather northeast–southwest in this part of the basin. The magnetic lineation is therefore interpreted to be of tectonic origin and superimposed on the sedimentary signal (the three axes of the ellipsoids of magnetic susceptibility are interdependent), and related to the formation of fold F1 in the basin. It implies an east–west direction of strain in the first stage of the basin inversion, in the late Early to early Middle Devonian. Such a direction of shortening is unlikely to generate the present-day morphology of the basin without the intervention of a second deformation (Figure 3a). From this, it can be concluded that the magnetic lineation was acquired in relation to fold F1, since the mean K_{MAX} axis tends to align with the fold axis, and passively rotated during F2 folding. This would imply that F1 and F2 are not coeval but F2 is of second generation relative to F1. Paleomagnetic results from locality CHU give an indication for a possible second phase of folding (see below).

The magnetic lineation K_{MAX} in the dykes intruding the top of the sampled section is very steep and well

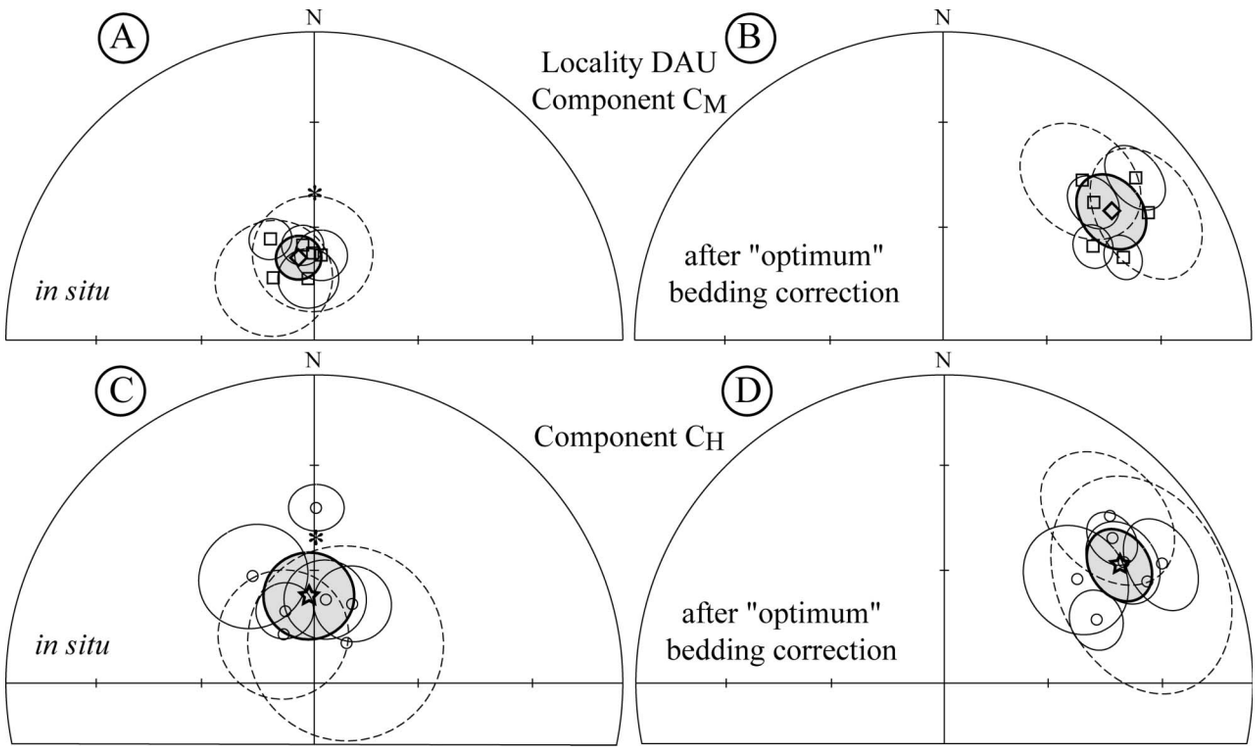


Figure 7 (a, b) Orientation of components obtained from the sandstone of the Mt Daubeny Formation (locality DAU). Overall mean direction (diamond with associated shaded cone at the 95% confidence level) calculated from site-mean directions (squares and associated α_{95}) for component DAU-C_M shown (a) *in situ* and (b) after optimum bedding correction. (c, d) Overall mean direction (star with associated shaded α_{95}) calculated from site-mean directions (circles and associated α_{95}) for component DAU-C_H shown (c) *in situ* and (d) after optimum bedding correction. The dashed confidence cones correspond to the two sites the closest to the conglomerate and where the grainsize is coarser. The asterisks show the direction of the present-day field *in situ*. Open symbols are negative inclination values (upper hemisphere); close symbols, positive inclination values (lower hemisphere). Equal-area polar projection.

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corresponds to a subvertical direction within the dykes plane (Figure 4c). K_{MIN} axes are distributed along a great circle and no preferred direction is recorded. It is simply interpreted as reflecting the intruding magmatic flow.

Magnetic mineralogy

The IRM, back-field and hysteresis curves show that all samples are dominated by high coercivity minerals (Figure 5a, b). These are predominantly goethite and hematite, as revealed by the thermomagnetic curve (Figure 5c) with Curie temperatures calculated at approximately 120°C and 670°C, respectively (Leonhardt 2006). However, thermomagnetic curves, IRM, and back-field curves indicate also the presence of a low-coercivity magnetite-like mineral, with Curie temperature calculated at ~580°C. In addition, all curves are characterised by a strong paramagnetic contribution.

Note that samples taken in the sandstone close to the dyke (at 2 and 5 cm: samples DAU.7–1 and DAU.7–2) show a much lower coercivity material content than samples taken further away (at 152 cm: sample DAU.7–7).

Paleomagnetic results

DEMAGNETISATION AND PRINCIPAL-COMPONENT ANALYSIS

Detailed thermal demagnetisation (26 steps) carried out on the sandstones of the Mt Daubeny Formation isolated two characteristic components in addition to a present-day field component. Goethite is presumed to be the magnetic carrier of the minor present-day field component as it is removed below 80 or 120°C. The present-day field component, however, is usually strongly deviated by a viscous or drilling-induced magnetisation. The lower temperature characteristic component, named component DAU-C_M, is fully demagnetised by 575°C, and the higher temperature characteristic component, named component DAU-C_H, is removed at 675°C. These unblocking temperatures, coupled with rock magnetic analyses, indicate that DAU-C_M is most probably carried by magnetite, and DAU-C_H, by hematite.

The two directions of magnetisation are usually distinct (Figure 6c), although in some cases distinction between components cannot be firmly established (Figure 6d). The overall mean direction DAU-C_H, however, resembles that of DAU-C_M (Figure 7). The *in situ* orientations are relatively close to the present-day field (asterisk in Figure 7), but all site mean directions are significantly distinct from it (i.e. no intersection with their confidence cone). It must be further noted that the two sites with the larger confidence cones (dashed cones in Figure 7) are situated at the base of the section (just above the conglomerate), where grainsize is coarser.

FOLD TESTS

An unequivocal fold test is difficult to obtain because the change in bedding orientation is small throughout

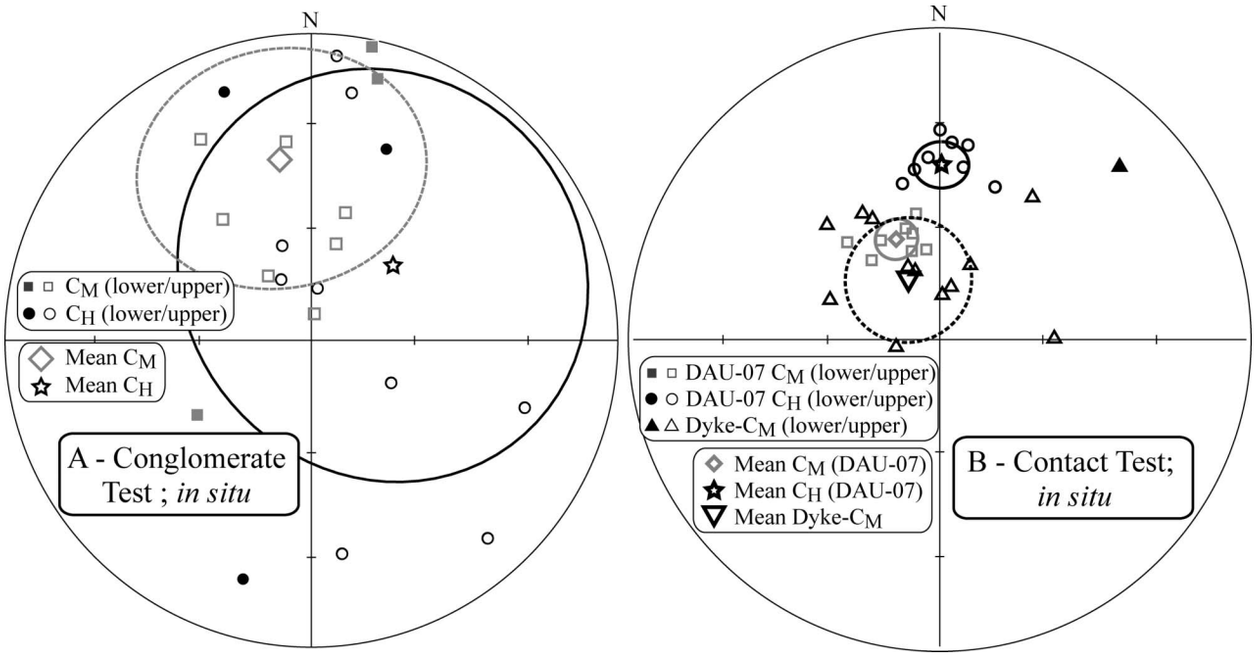


Figure 8 (a) Conglomerate test: single specimen orientations for components C_M (grey squares) are scattered but not randomly distributed as they tend to group around their mean direction (diamond with associated grey α_{95}). On the contrary, orientations of C_H (circles) show a uniform distribution at the 95% confidence level and their mean direction (star and black α_{95}) has no significance. (b) Contact test: single specimen orientations for C_M from the dykes (triangles) are scattered but their mean direction (inverse triangle with associated dashed α_{95}) is not distinguishable from the mean direction (diamond with associated grey α_{95}) obtained from components DAU-C_M (squares) of the surrounding sandstone. On the contrary, it is significantly distinct from the mean direction (star and black α_{95}) obtained from components DAU-C_H (circles) of the sandstone. Symbols as in Figure 7.

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locality DAU. Therefore, it has been chosen to show the outcomes of many fold tests for comparison—the classic fold test of McElhinny (1964), the algorithm of McFadden (1990), the parametric simulation of Watson & Enkin (1993), the bootstrap fold test of Tauxe (1998), and the DC-Slope test of Enkin (2003). Note that, although the classic fold test of McElhinny (1964) is commonly viewed as more stringent than others, the test is, particularly in this case, mathematically not strictly valid (not Fisher-distributed). The McFadden fold test is equally not valid in theory, because it is designed for specific distribution of sites on both limbs of a fold.

Site mean directions from component DAU-C_M are plotted on a stereogram *in situ* and after bedding correction (Figure 7a, b; Table 1). The overall mean direction *in situ* for the locality is: D.349°/I.-68° ($\alpha_{95} = 5.9°/\kappa = 131.6$), and yields after optimum bedding correction: D.053°/I.-32° ($\alpha_{95} = 9.4°/\kappa = 52.1$). The classic fold test (McElhinny 1964) has a κ-ratio = 0.41 and suggests a post-folding remagnetisation. Furthermore, the *in situ* orientation corresponds to a segment of the APW path for Australia (Early Cretaceous; see below).

The overall mean direction for the component DAU-C_H (Figure 7c, d; Table 1) is: D.357°/I.-67° ($\alpha_{95} = 11.7°/\kappa = 27.6$) *in situ*, and D.056°/I.-32° ($\alpha_{95} = 8.9°/\kappa = 47.0$) after optimum bedding correction. Although a bias in statistics is introduced due to the complex unfolding procedures, the classic fold test (McElhinny 1964) does not yield a positive result at the 95% confidence level. Nevertheless, the data pass the test at the 81.4% confidence level when performed on site-mean directions, and at the 93.2% confidence level when using single sample orientations. The algorithm of McFadden (1990) yields a positive fold test at the 95% confidence level. In addition, both the DC-Slope fold test (Enkin 2003) and the bootstrap analysis (Tauxe 1998) are positive at the 95% confidence level.

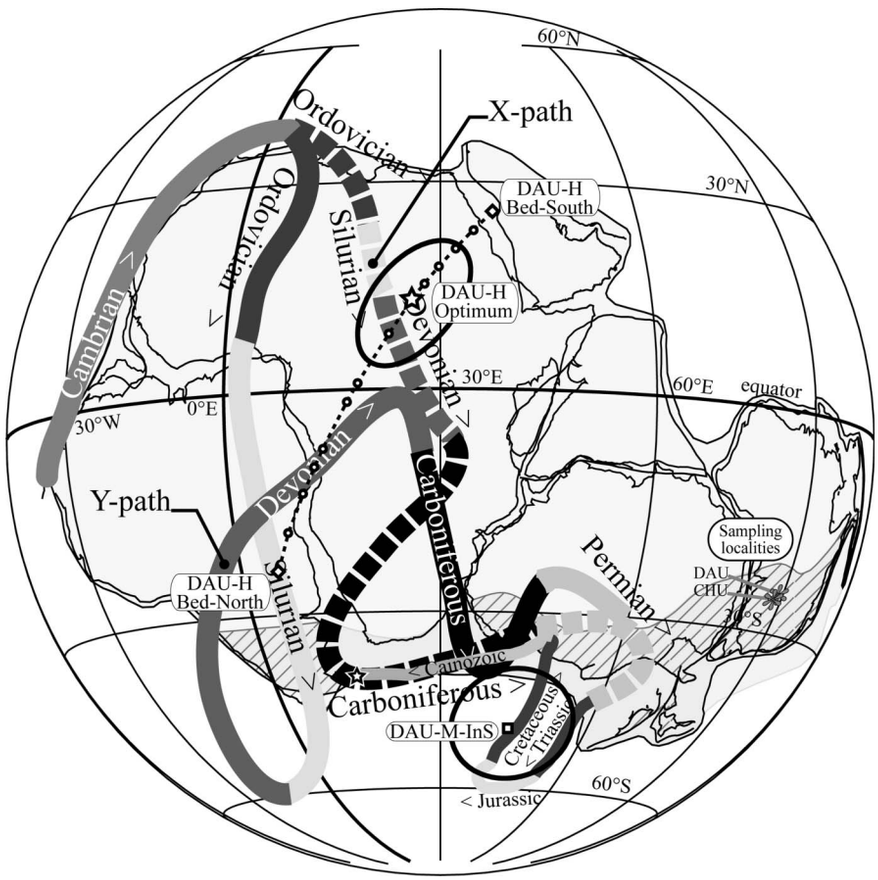


Figure 9 Pole positions obtained from the Mt Daubeny Formation in the Mt Daubeny Basin (locality DAU), and plotted together with the X- and Y-paths presented in Figure 1. The pole corresponding to the overall mean direction obtained after optimum bedding correction is noted DAU-H Optimum (star with the associated confidence ellipse), and well match the Siluro-Devonian segment of the X-APW path of Bachtadse & Briden (1991). For comparison, the poles corresponding to all bedding corrections tested (5° steps; see text) are also shown (circles). The poles corresponding to the two end-member solutions for the bedding correction are represented with diamonds, and labelled DAU-H-Bed South when the northern part of the basin is rotated to align the fold axis F1 of the southern part (case in Figure 3 C_3), and DAU-H-Bed North for the contrary (case of Figure 3 C_1). DAU-M-InS represents the pole position of the direction of magnetisation *in situ* carried by magnetite. It is consistent with the Early Cretaceous segment of the APW path for Australia.

Because the superiority of the optimum bedding correction is tenuous relative to all other possibilities (Figure 3), fold tests have also performed with the other configurations. Positive fold tests at the 95% confidence level are also obtained with the DC-Slope test of Enkin (2003) and the parametric simulation of Watson & Enkin (1993) with some of the corrections around the optimum solution. Note that the overall mean directions for the two DAU-C_H end-members are D.013°/I.- 32 ($\alpha_{95} = 10.5°/\kappa = 33.7$) when the fold axis of F1 is rotated in the position of the northern part of the basin (Figure 3 C_1), and D.074°/I.-32° ($\alpha_{95} = 9.3°/\kappa = 43.3$) when the fold axis of F1 is rotated in the position of the southern part of the basin (Figure 3 C_3).

When fold tests are conducted separately on F1 and F2, the classic fold test (McElhinny 1964) yields a quasi-unchanged κ-ratio for F1, and indicates a non-significant improvement for F2. However, not one of the various tests performed are significantly negative, suggesting that data require correction from the two folds.

CONGLOMERATE TEST

A true conglomerate test would theoretically consist in sampling pebbles derived from the Mt Daubeny Formation. This is not possible here, since the conglomerate lies near the base of the formation and recycles Cambrian rocks. The sampled conglomerate comprises two lithologically distinct pebbles (volcanic and sandstone) encased within the sandy matrix of the Mt Daubeny Formation. The difference in the petrology of the pebbles corresponds to a difference in paleomagnetic behaviour of the samples.

After removal of the viscous or drilling-induced magnetisation, however, the volcanic pebbles show one component of magnetisation, whereas the sandstone pebbles show two directions. In the sandstone pebbles, the first component is carried by magnetite and is compared with component DAU-C_M identified in the red sandstone. The second is carried by hematite and is compared with component DAU-C_H. In the volcanic pebbles, however, both magnetite and hematite carry the same direction of magnetisation in a specimen. This component is compared with component DAU-C_H.

Orientations of C_H obtained from pebbles are randomly distributed. The Rayleigh (Rayleigh 1919) and Beran/Giné statistics (Beran 1968; Giné 1975) confirm uniformity at the 95% confidence level.

Conversely, orientations of C_M seem to show a slight grouping about the mean direction and statistical tests reject uniformity (Figure 8a, shown *in situ* only, as the distribution is the same after bedding correction because the stratigraphy is monoclinal within the site: simple tilting does not modify data distribution).

As the magnetic carriers are similar, one can conclude, at least, that no important thermal remagnetisation affected this zone. From this point of view, the conglomerate test may be regarded as a negative test concerning the components C_M, but a positive test concerning C_H.

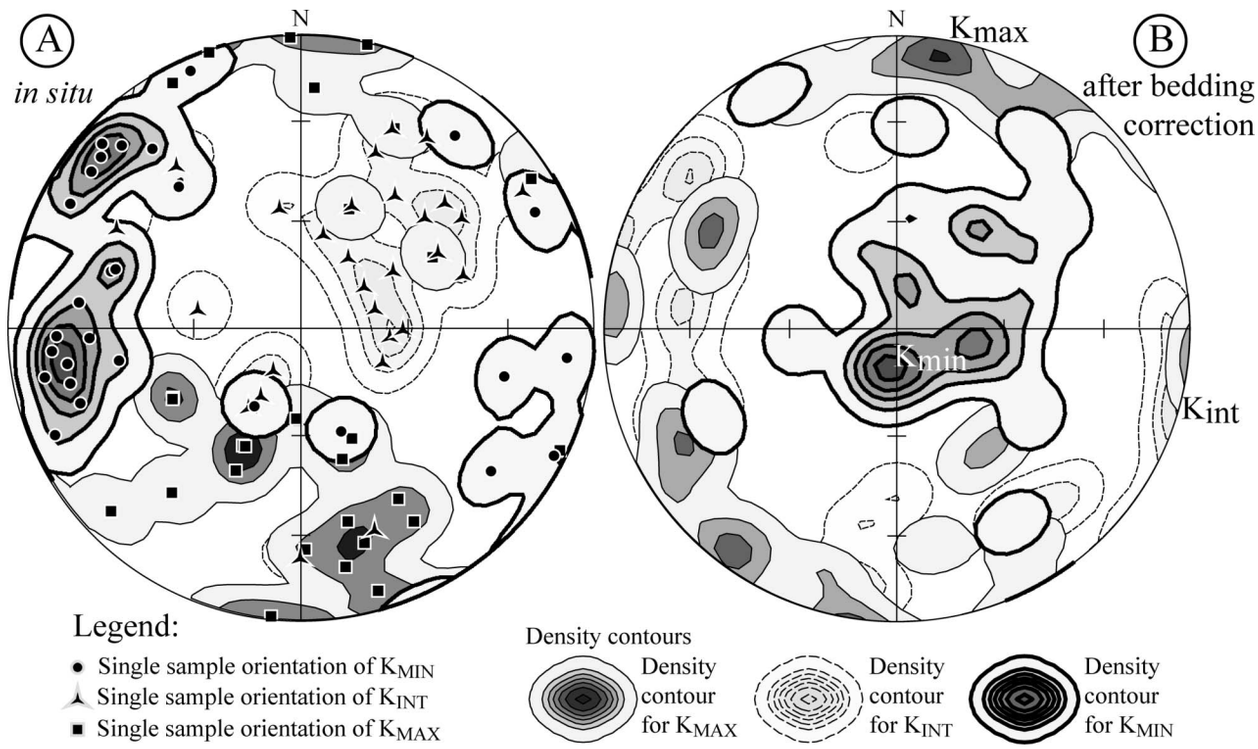


Figure 10 Orientations of the principal axes of magnetic susceptibility for the Mt Daubeny Formation from the Churinga Station (locality CHU), data points and density contours: (a) *in situ*; and (b) after bedding correction (for sake of clarity, only density contours are shown). Symbols as in Figure 4.

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CONTACT TEST

Three dykes have been drilled. One is ~1.5 m-wide, and the other two are narrower (~50 cm). Macroscopically, the presence of chlorite attests to some alteration. The contact between the dykes and the red sandstone of the Mt Daubeny Formation is sharp, and the first couple of centimetres in the sandstone are pale, almost white. Ten cores have been drilled in the sandstones (site DAU-07) for the contact test at 2, 5, 10, 20, 32, 74, 152, 238, 310 and 412 cm from the edge of one dyke (~60 cm thick). Note that sampling was possible only in one direction away from the dyke.

The dykes have a strong viscous magnetisation up to 80–120°C, but contain basically one component of magnetisation with unblocking temperatures reaching 550°C (Figure 6k–n). This magnetisation is carried by some low-titanium titano-magnetite and can be named component C_M. Orientations yielded by this component are very scattered (triangles in Figure 8b, shown *in situ*), probably due to the alteration visible in these rocks.

When compared with directions from the sandstone (Figure 8b), the mean direction obtained from the dykes (inverse triangle) is not statistically different from the direction of the component DAU-C_M (diamond) at site 07, which lies within the confidence cone (dashed a_{95}). Conversely, it is distinct from the direction yielded by the component DAU-C_H (star) of the sandstones (site 07), as confidence cones from the two directions are clearly separated.

However, there is no deviation in magnetisation in the sandstone when approaching the dyke, as expected for a textbook contact test. The sandstone becomes paler (almost white) close to the dyke suggesting hematite vanishes. This might be the reason why the deviation expected is not observed. Yet, the directions from the high-temperature component DAU-C_H identified in the sandstone are statistically distinct from directions of magnetisation in the dykes, and hence,

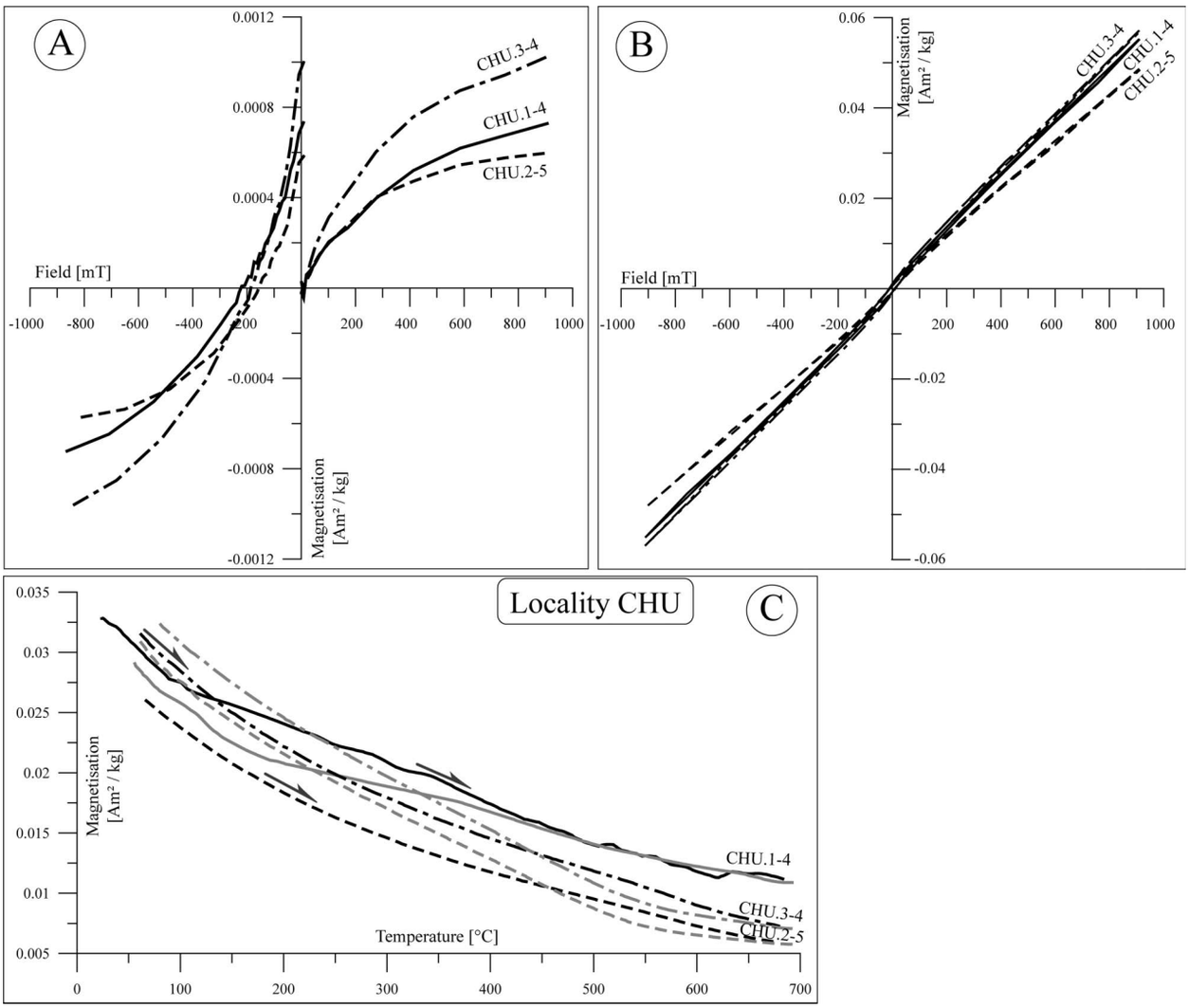


Figure 11 Examples of magnetic mineralogy measurements for locality CHU. (a) Back-field (left) and IRM (right) curves; (b) hysteresis loop; and (c) thermomagnetic curves ($B = 530$ mT), with black lines corresponding to heating cycles, grey lines to cooling cycles. CHU.1–4, solid line; CHU.2–5, dashed line; CHU.3–4, dotted-dashed line.

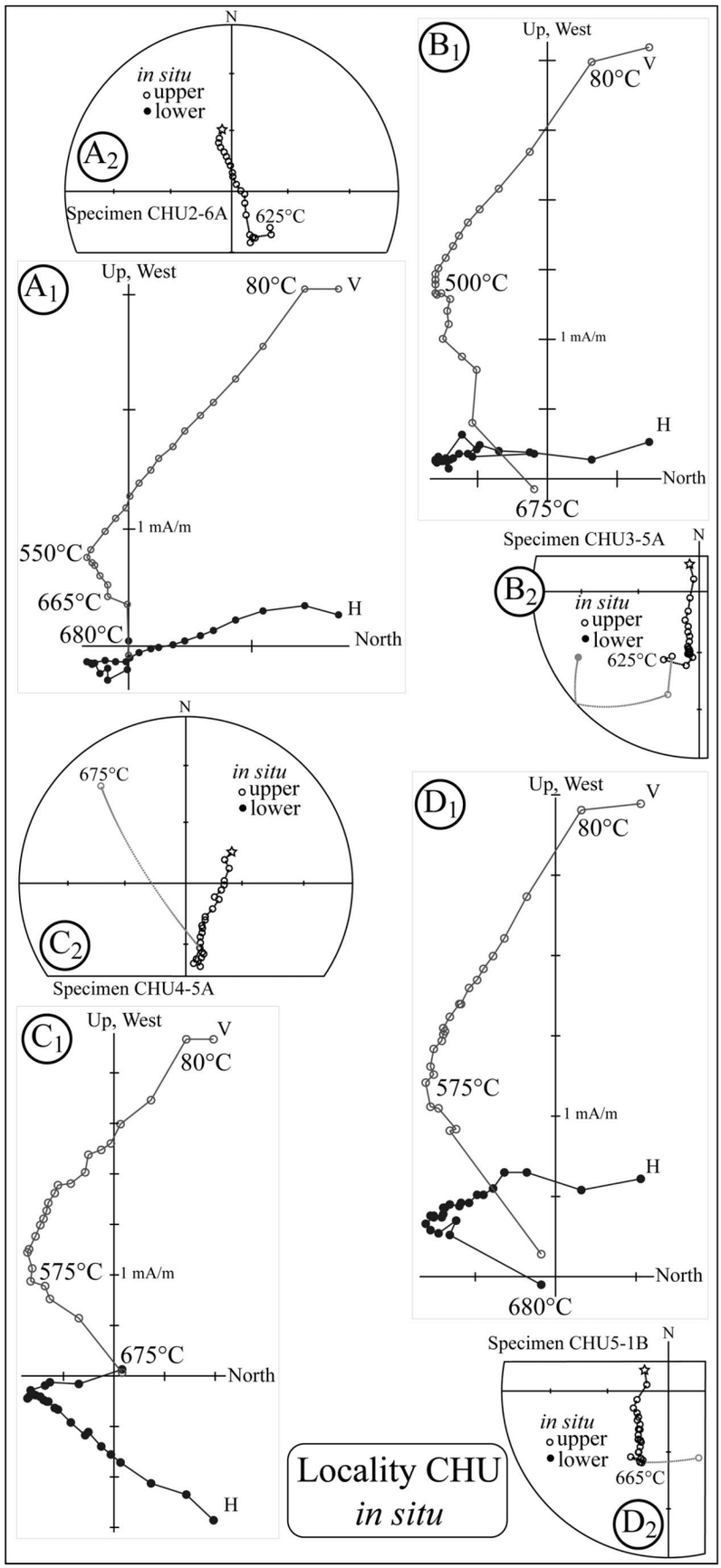


Figure 12 Examples of paleomagnetic results obtained by thermal demagnetisation from the Mt Daubeny Formation at Churinga Station (locality CHU). (a) CHU.2–6; NRM at room temperature: 3.425 mA/m: A_1, orthogonal projection; A_2, stereographic projection. (b) CHU.3–5; NRM at room temperature: 3.394 mA/m. (c) CHU.4–5; NRM at room temperature: 3.752 mA/m. (d) CHU.5–1; NRM at room temperature: 3.062 mA/m. Legend as in Figure 6.

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the contact test may be considered by some as positive.

Interpretation and discussion

The mid-temperature components DAU-C_M from the Mt Daubeny sandstone have negative fold tests. Components carried by magnetite in the conglomerate are not randomly distributed, and the direction of DAU-C_M is statistically indistinguishable from the mean direction obtained from the dykes. This is clearly indicative of a remagnetisation event. Moreover, the corresponding paleopole to the *in situ* orientation of DAU-C_M is: $\varphi = 162.26^\circ E / \lambda = 68.57^\circ S$ ($dp = 8.20^\circ /$

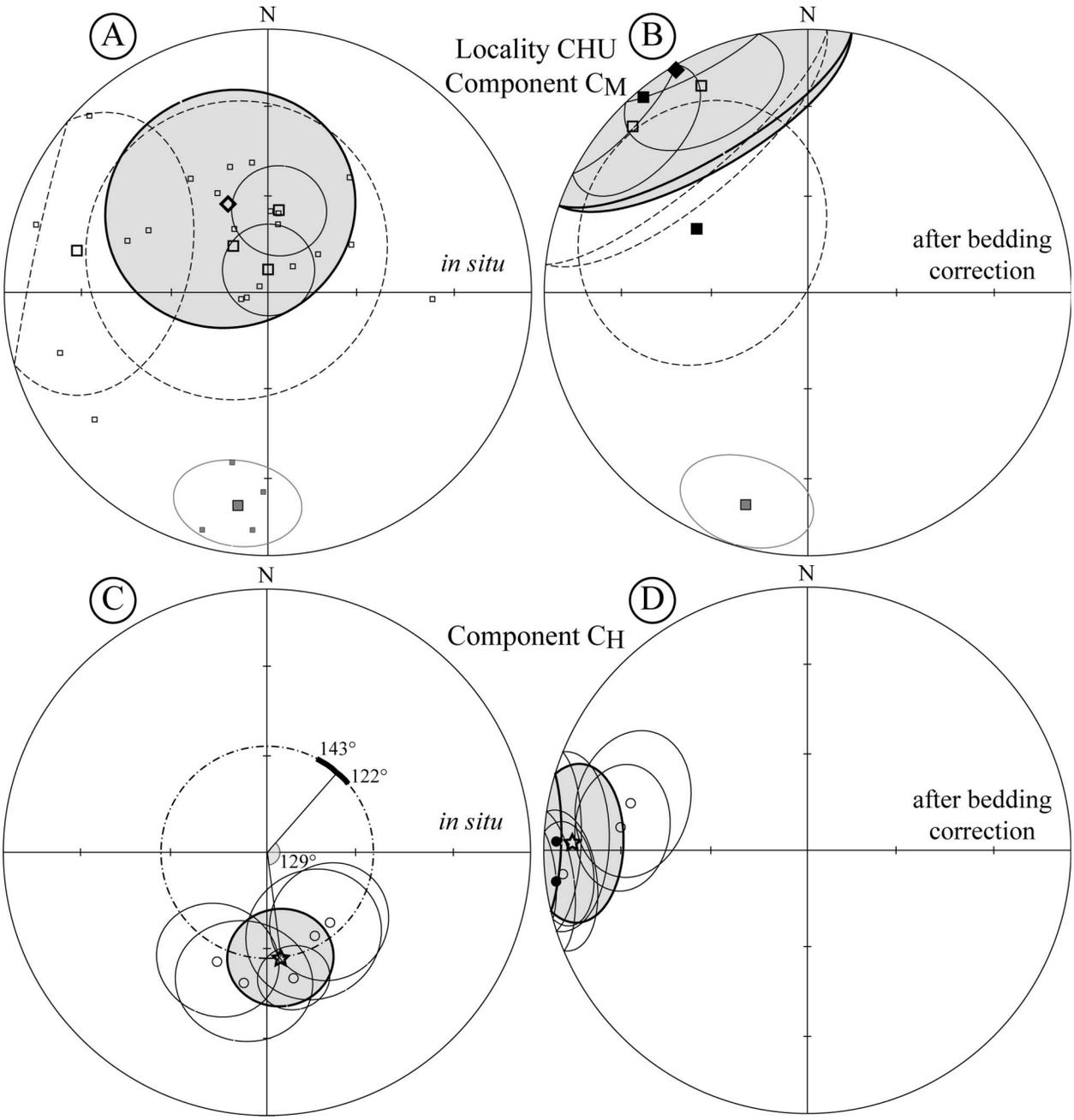


Figure 13 Orientation of components obtained from the Mt Daubeny Formation at Churinga Station (locality CHU). (a, b) Overall mean direction (diamond with associated shaded cone at the 95% confidence level) calculated from site-mean directions (squares and associated α_{95}) for component CHU-C_M shown: (a) *in situ*; and (b) after bedding correction. Single specimen directions are also presented *in situ* (small squares); it shows that directions may be relatively clustered per site (one example corresponds to the site CHU-03 shown in grey). Other sites may be very scattered and the corresponding confidence cone, very large (i.e. $>30^\circ$; shown with dashed lines). (c, d) Overall mean direction (star with associated shaded α_{95}) calculated from site-mean directions (circles and associated α_{95}) for component CHU-C_H shown: (c) *in situ*; and (d) after bedding correction. A $\sim 130^\circ$ rotation around a vertical axis (dotted-dashed circle) can match the Late Devonian–Early Carboniferous segment of many published APW paths for Gondwana. Symbols as in Figure 7.

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dm = 9.80°: Australian coordinates). Although it falls near (i.e. the confidence cone overlaps) the Triassic–Jurassic segments of the X- and Y-APW paths (Figure 9), the paleopole lies most closely to the Early Cretaceous segment of the APW path for Australia (Figure 9; Table 2).

The pole, therefore, may be interpreted as reflecting an Early Cretaceous event, which age could be linked to other dykes and pipe-like intrusions of alkaline basalts to the northeast of the Broken Hill area (Stevens *et al.* 2000). Fluid migration associated with the magmatic event, in particular, could explain why this magnetisation (C_M) is present in the pebbles of sandstones in the conglomerate if the fluid percolated into them but not in the more impervious volcanic ones. Moreover, the association of pre-existing and fluid-deposited magnetite in the pebbles of sandstone can cause the observed scattering in orientations of C_M (squares in Figure 8a). In addition, a fluid associated with the magmatic event may have circulated through the sandstone depositing magnetite throughout the formation. However, the fading of the red colour of the sandstone could be understood as the results of a preferred path for the fluid along the dykes, leading to their alteration and the formation of the observed chlorite.

Concerning component DAU-C_H, the interpretation of the age of magnetisation is more difficult. On the one hand, various fold tests are positive at the 95% confidence level. Moreover, the same bedding correction worsens data clustering for C_M while it improves it for C_H. The uniform distribution of C_H is accepted at the 95% confidence level in the conglomerate test, and the contact test shows a statistically distinct orientation of components C_M from the dykes relative to component DAU-C_H of the surrounding redbeds. Taken together, and given that the folding occurred soon after deposition, one may think that the three types of test point to the same conclusion: DAU-C_H can be considered pre-folding and, therefore, primary in origin. On the other hand, none of the available paleomagnetic tests are textbook examples, and each test can be questioned. The most problematic result is the resemblance of the overall mean direction of DAU-C_H with that of DAU-C_M *in situ*.

In addition, even if the magnetisation is primary, the position of the corresponding paleopole remains unclear, given the uncertainty concerning the exact

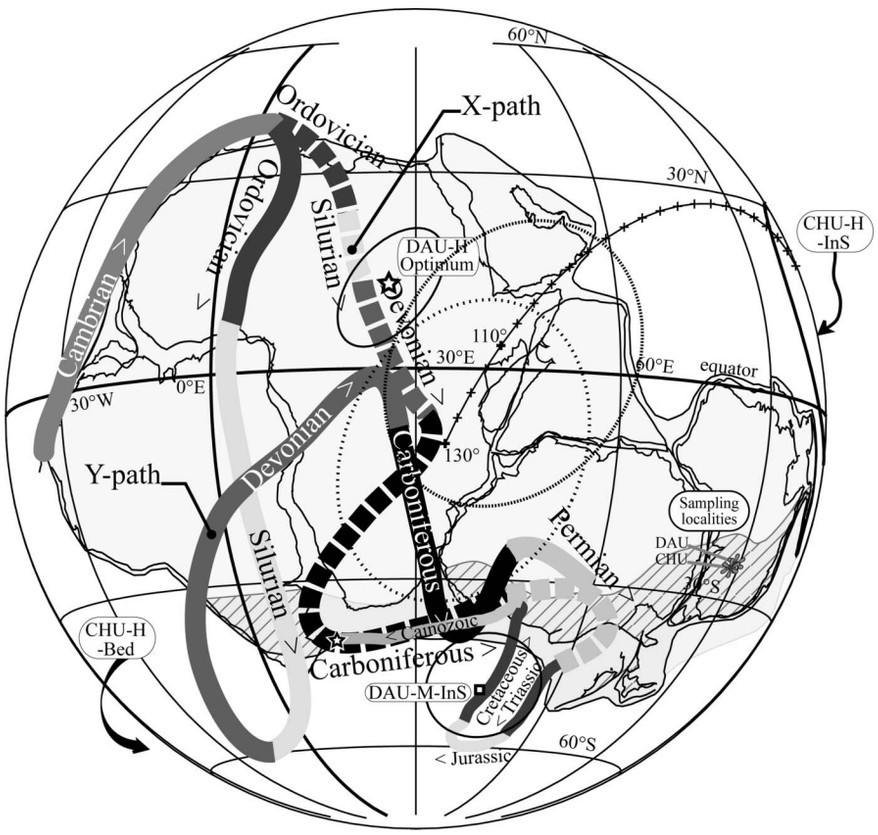


Figure 14 Pole positions obtained from the Mt Daubeny Formation at the Churinga Station (locality CHU). Both poles corresponding to the overall mean direction obtained *in situ* (CHU-H-InS) and after bedding correction (CHU-H-Bed) plot far away from any expected position for Gondwana (even if the alternate polarities are considered), and lie on the back side of the Earth relative to the X- & Y-type of APW paths. CHU-H-InS can match the Late Devonian–Early Carboniferous segment of the APW path for Gondwana only after at least a 110° counter-clockwise rotation, where the confidence cone overlaps this segment. However, the pole CHU-H-InS better corresponds to the Early Carboniferous segment after a 130° rotation. Each cross represents 5° rotation around a vertical axis situated at the sampling locality. The poles obtained for component C_M (DAU-M-InS) and for C_H (DAU-H Optimum) at locality DAU are shown for comparison.

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bedding correction to apply. Consequently, all solutions calculated every 5° steps between the two end-members for the envisaged bedding corrections are plotted in Figure 9, together with the paleopole determined from DAU-C_H after optimum bedding correction, which is $\varphi = 051.23°E/\lambda = 37.61°S$ ($dp = 5.63°/dm = 10.01°$: Australian coordinates). Note that the pole DAU-H-Optimum is compatible with the Siluro-Devonian segment of the X-type of APW path. Conversely, none of the solutions envisaged are in agreement with the Siluro-Devonian segment of the Y-type of APW path (Figure 9).

The question arises as to whether the complex bedding correction should be applied or whether rotation around a vertical axis should be inferred from paleomagnetic results. Second-order fold F2 must be taken into account to perform fold tests, and correcting from the second-order fold F2 in order to have a planar limb remained fixed in the position as is in the northern part of the basin (Figure 3 C_1) is actually the option corresponding to pole DAU-H_Bed-North (Figure 9). Moreover, the APW path for Gondwana is not clearly established. It is not possible in the present state of knowledge, therefore, to infer the amount of rotation around the vertical axis. Thus, we should take advantage of the structural information of the Mt Daubeny Basin to discuss and propose a best or optimum solution.

LOCALITY CHU: CHURINGA STATION

Churinga Station (CHU) is situated ~80 km south of the locality DAU (Figure 2), in a structurally different and independent syncline, affected by metre-scale second-order folds (see above). Rocks at the locality CHU are poorly exposed in a dried creek bed.

Anisotropy of magnetic susceptibility

Most samples of locality CHU display a very low degree of anisotropy ($P_J = 1.02 \pm 0.01$) and the ellipsoid of magnetic susceptibility tends to have an oblate shape ($T = 0.44 \pm 0.33$). Internal strain is, therefore, interpreted as negligible throughout this sampling area, and no lightning effects are expected given the value of the Königsberger ratio ($Q_K = 0.26 \pm 0.10$).

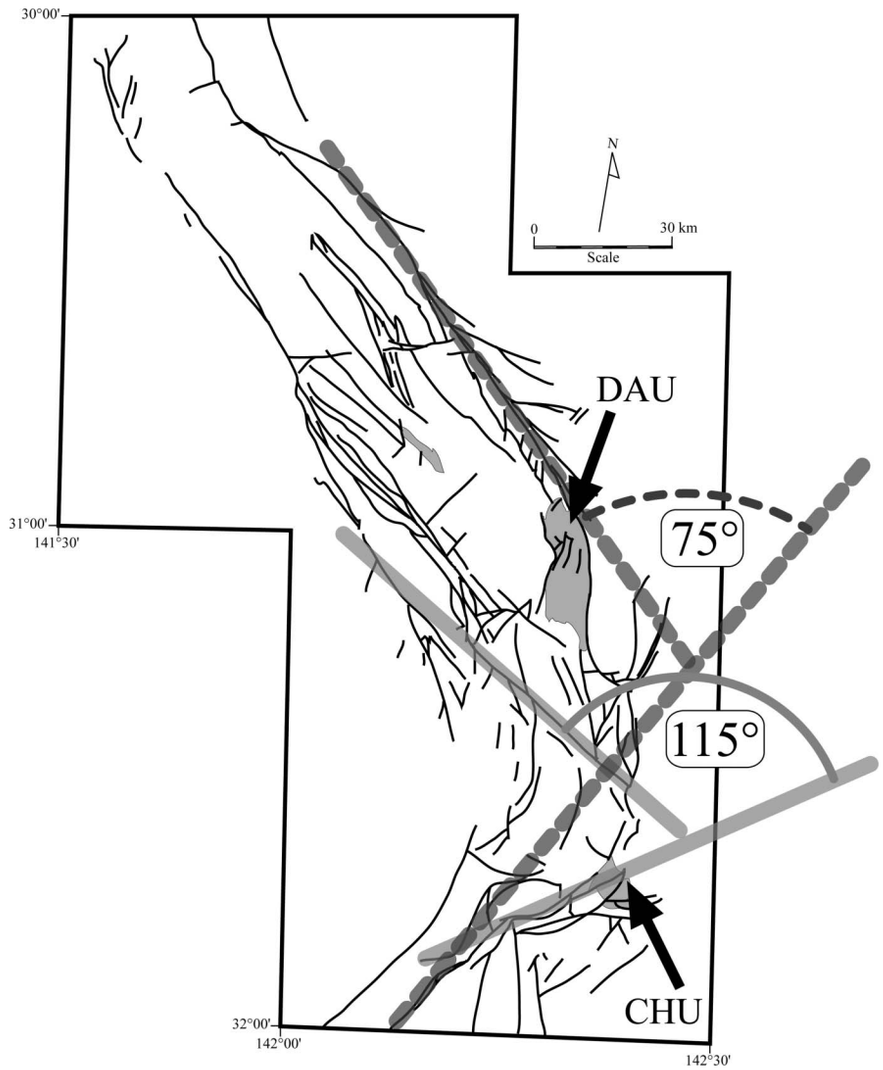


Figure 15 Same map as Figure 2c, showing only the main faults (solid lines) and the Mt Daubeny Formation (grey) in order to highlight the change in structural trend at the Grasmere Knee Zone. Although the change of structural trend is ~75° to 90° at regional scale, the amount of change appears to be locally much higher—115°. It corresponds to the area where locality CHU is situated, and where a rotation of same order is proposed.

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The directions of the principal axes of magnetic susceptibility (Figure 10) show two clusters of K_{MIN} axes *in situ* corresponding to the two limbs of the second-order folds present, whereas after bedding correction the clusters merge and are better grouped about the vertical. This is indicative of a magnetic fabric still linked with the sedimentary plane, in good agreement with the oblate shape of the ellipsoid of magnetic susceptibility. No magnetic lineation can be clearly determined although the maximum density is directed approximately north–south after bedding correction.

Magnetic mineralogy

Magnetic mineralogy analysis is obscured by the strong paramagnetic signal (Figure 11), and is not dealt with in detail here. By subtracting the paramagnetic contribution and, in particular, by using a fitting function (Leonhardt 2006), the presence of two main phases could be characterised in all samples, and are interpreted as magnetite and hematite. Goethite is also present, but in lesser quantity than at locality DAU.

Paleomagnetic results

DEMAGNETISATION

The demagnetisation process clearly shows two components after removal of a mainly viscous or drilling-induced magnetisation (Figure 12): a component CHU-C_M up to 550°C most probably carried by magnetite, and a component CHU-C_H from 550 to 680°C, carried by hematite, corroborating the identification from the rock magnetic measurements. Component CHU-C_H is not always well isolated due to acquisition of laboratory-induced magnetisation during the heating process, which obscures the natural remanent magnetisation (Figure 12c), although the two components are always clearly distinct from one another.

FOLD TESTS

Site-mean directions for CHU-C_M are shown together with single specimen components (Figure 13a), because orientations per site appear fairly clustered (for example, site CHU-03 shown in grey around 188°/19° in Figure 13a), except for two sites where confidence cones (shown with dashed line) are extremely large. Conversely, the overall mean direction has a very large confidence cone, because between-site mean directions are very scattered, both *in situ* (Figure 13a) with an overall mean direction D.336°/I.–60° ($N = 27$; $\alpha_{95} = 38.2°/\kappa = 5.0$), and after bedding correction (Figure 13b): D.329°/I. + 02° ($\alpha_{95} = 40.1°/\kappa = 4.6$). The κ-ratio equals 0.92, and none of the applied fold tests is significant.

By contrast, the site mean directions calculated for the components CHU-C_H are much better grouped than CHU-C_M (Figure 13c, d; Table 3), and the overall mean direction for the locality is *in situ*: D.173°/I.–57° ($\alpha_{95} = 15.7°/\kappa = 24.7$) and after bedding correction: D.272°/I.–12° ($\alpha_{95} = 19.1°/\kappa = 17.1$). However, the parametric simulation fold test is negative at the 95% confidence level, whereas the other fold tests applied are not significant. The grouping, in any case, worsens after bedding correction, suggesting the acquisition of a post-folding magnetisation.

Interpretation and discussion

No clear explanation has been found to interpret the behaviour of component CHU-C_M. However, the AMS shows that internal strain is not strong enough to be responsible for deviation of the remanence. One can only speculate that several directions of magnetisation are removed simultaneously. In spite of its large confidence cone, the overall mean direction remains consistent with the direction of remagnetisation (DAU-C_M) observed at locality DAU.

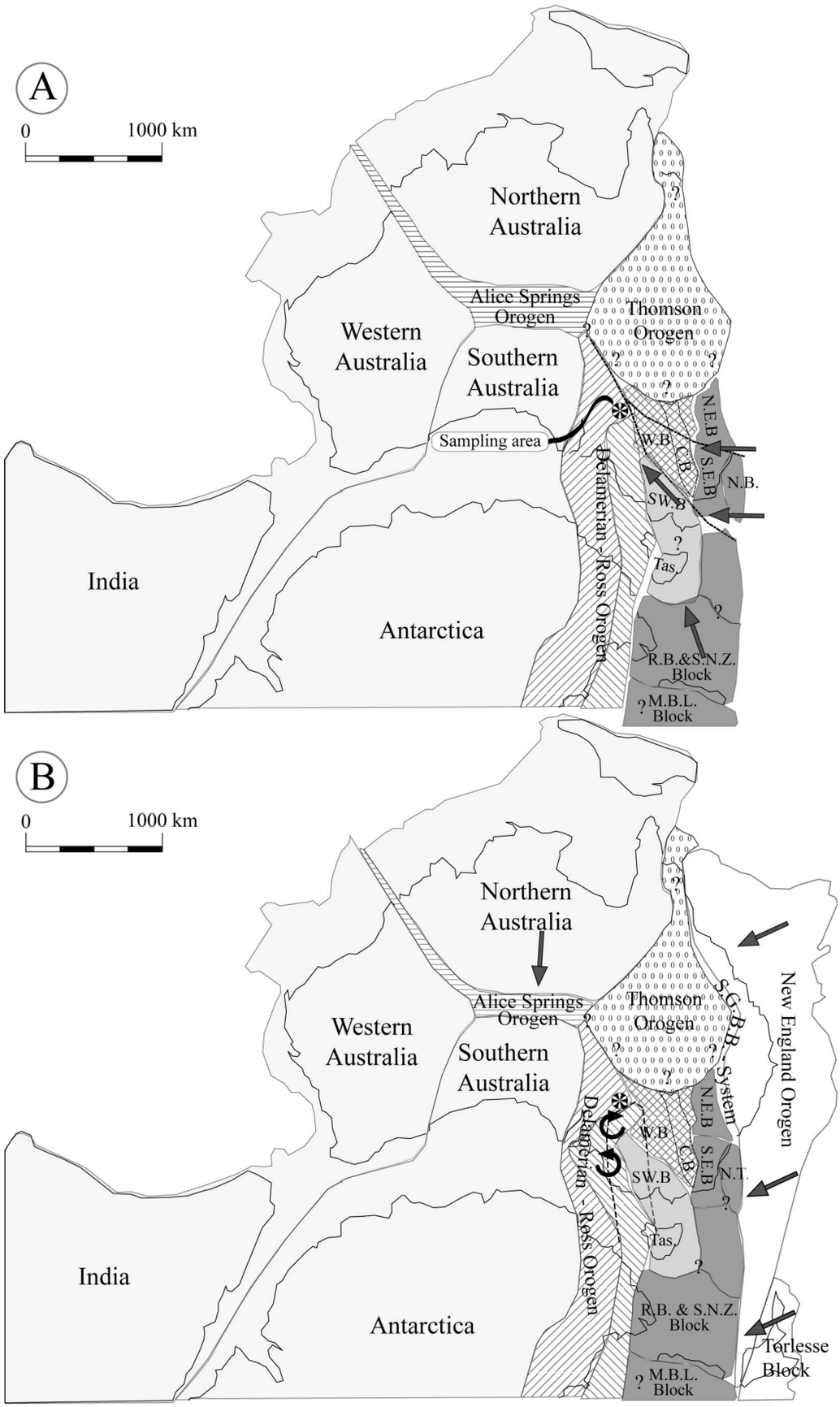
This behaviour could be understood, therefore, as partial remagnetisation affecting certain sites more than others. The possible Mesozoic overprint observed in the red sandstone of the Mt Daubeny Basin (locality DAU) might be also present here, and demagnetised together with an unknown north–south direction (site shown in grey in Figure 13). This result, however, cannot be further used as no reliable paleomagnetic direction can be determined.

The paleopoles from locality CHU corresponding to the *in situ* direction, and that corresponding to the direction after bedding correction (Table 4) for CHU-C_H, do not fall close to any Siluro-Devonian or younger segments proposed for the APW path for Gondwanan-Australia, even when the alternate polarities are considered. It is also different from the direction obtained from the Mt Daubeny Formation at locality DAU. The site-mean directions are better clustered *in situ* than after bedding correction according to the different fold tests.

Figure 16 Tentative model explaining the occurrence of block rotation in the Tasmanides. (a) A block corresponding roughly to the Southwestern Subprovince of the Lachlan Orogen indents the region between the Central Subprovince of the Lachlan Orogen and the Delamerian Orogen in the Devonian. (b) The movement is still active in the Late Devonian to Early Carboniferous, triggering the clockwise rotation of the southern part of the Broken Hill area and possibly sinistral strike-slip movement along the Koonenberry Fault. A counter-clockwise rotation should thus be observed in western Victoria (rounded arrows). The dashed line recalls the position of the Eastern Delamerian Orogen as shown in (a). At this time the Alice Springs Orogen in central Australia and the New England Orogen along the eastern margin form. In this model, note that the term 'block' is preferred to the better defined terms 'terranes' or 'subprovinces.' S.W.B., Southwestern block; C.B., Central block; N.E.B., Northern Eastern block; S.E.B., Southern Eastern block; N.T., Narooma block; Tas., Tasmania; R.B. & S.N.Z., Robertson Bay and Southern New Zealand block; M.B.L., Marie Byrd Land block. Paleogeographic framework based on models of Sutherland (1999) and Müller *et al.* (2000).

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The locality is thus regarded as remagnetised, but it must be considered that the *in situ* direction of remagnetisation has been affected by a post-folding rotation. Because of its inclination, an approximately 130° rotation of the overall mean direction around a vertical axis can bring this direction close to a Late Devonian to mid-Carboniferous segment of both types of APW path presented for Gondwana (Figures 13c, 14).

CONCLUSIONS AND SPECULATIONS

Two components of magnetisation have been interpreted from the thermal stepwise demagnetisation carried out on the Siluro-Devonian sandstone of the Mt Daubeny Formation (locality DAU). Component C_M, carried by magnetite, is correlated with a Mesozoic, probably Early Cretaceous, remagnetisation event, and is consistent, in particular, with fluid migration that might have been associated with the intrusions of alkaline basalts to the northeast of the Broken Hill area.

The age of acquisition of component C_H could not be firmly established and remains subject to caution. The most problematic point concerns the resemblance between the overall mean directions of component C_H and C_M. If components C_M and C_H both represent the same remagnetisation event, why are the components C_H and C_M in most samples in different orientations? And why does the same bedding correction improve data clustering for C_H and worsen it for C_M? The dyke test performed shows no deviation of the magnetisation when approaching the dyke. But, if the dyke is altered and remagnetised by a percolating fluid, no deviation is to be expected, and the significant difference in direction of magnetisation between the dyke and the surrounding sandstone favours an older age of acquisition for C_H. However, in this case the dyke test is no longer a contact test. The conglomerate test indicates a uniform distribution of C_H in the pebbles, but concluding a positive test would be questionable since pebbles are not derived from the Mt Daubeny Formation. Finally, the complexity of folding renders suspicious the results of fold tests, although most of them indicate a pre-folding (primary) origin of the magnetisation C_H at the 95% confidence level.

Nevertheless, if one accepts the primary origin of component C_H and agrees with the optimum bedding correction solution, three main and interdependent conclusions can be drawn: (i) an X-type of APW path for Gondwanan-Australia should be favoured; (ii) according to the segment of the X-APW path, the age of magnetisation is Siluro-Devonian, in agreement with the interpretation of the primary origin of the magnetisation; and (iii) since none of the possible paleopoles determined here could match the Siluro-Devonian segment of the Y-type path, the assumption that terranes from eastern New South Wales, on which the Siluro-Devonian Y-APW path is anchored, are (par-) autochthonous is erroneous.

The second significant finding of this study is that the pole of locality CHU does not correspond to any Siluro-Devonian or younger segments of any APW paths (of both X- and Y-types). One hypothesis is, therefore, to propose that block rotation occurred at some point after this magnetisation.

Regardless of confidence cones, this rotation best coincides with most APW paths of both types after a ~130° counter-clockwise rotation (Figures 13c, 14). Assuming the age and position of these reference paths to be correct, the age of remagnetisation at locality CHU would be Late Devonian to mid-Carboniferous, and the rotation would be after this time. It is unlikely that such a rotation would be younger than mid-Carboniferous, but a Late Devonian–Early Carboniferous age is consistent with major movements on the Koonenberry Fault and the effects of the Kanimblan–Alice Springs Orogeny (Mills & David 2004). Moreover, given the confidence cones of both the pole from locality CHU and reference paths, it is possible to relate the observed deviation in direction with the change in cleavage and fault trend marking the Grasmere Knee Zone. At a regional scale, this change is in order of ~75–90°, but at more local scale, where locality CHU is situated, it exceeds ~115° (Figure 15). These observations suggest that the southern part of the Broken Hill area has been rotated relative to the north.

One possible model to synthesise and speculate on the obtained results can, therefore, be summarised as follows.

The magnetisation carried by hematite (C_H) at locality DAU might have been acquired when the rock formed in the Siluro-Devonian. An east–west direction of shortening would have triggered the inversion of the Mt Daubeny Basin and its folding (F1) in the late Early to early Middle Devonian, with acquisition of a magnetic lineation (K_{MAX}) subparallel to the fold axis. The relationship between the direction of the pole to the magnetic foliation (K_{MIN}), which is usually much more stable than the lineation in sediments, and the pole to the bedding plane could have been preserved. Following the model of Glen *et al.* (1992), a block approximately corresponding to the Southwestern Subprovince of the Lachlan Orogen would have indented southeastern Australia between the Central Subprovince of the Lachlan Orogen and the Delamerian Orogen in the Middle to Late Devonian (Tabberabberan event) (Figure 16a). This tectonic event could have caused the Late Devonian to Early Carboniferous remagnetisation (C_H) observed at locality CHU. The Southwestern Block would have then continued moving north-northwest, perhaps until the Early Carboniferous, triggering clockwise rotation of the southern part of the Broken Hill area and the formation of the Grasmere Knee Zone together with the second phase of folding (F2) in the Mt Daubeny Basin (Figure 16b).

This hypothesis, reminiscent of the Powell *et al.* (1985) megakinking proposal for the coastal part (east) of the Lachlan Orogen, can be tested by finding counter-clockwise rotation in western Victoria, since the indenting Southwestern Block should have produced opposite rotation in the south (rounded arrows in Figure 16b).

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